Data and Artificial Intelligence
Cyber Shujaa Program

Week 4 Assignment
Business Intelligence with Power BI (Hotel Management)

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Introduction

The objective of this assignment was to develop hands-on experience in **Business Intelligence (BI)** using **Microsoft Power BI** by analyzing a real-world **Hotel Management dataset**. The project focused on transforming raw hotel data into meaningful insights that support data-driven decision-making in the hospitality industry.
The analysis involved understanding the hotel business context, loading and transforming multiple datasets (such as dim_date, dim_rooms, and fact tables), building a **star schema data model**, creating **DAX measures**, and designing an interactive dashboard. The final report was published to the **Power BI Service (cloud)** and documented as part of a professional analytics portfolio.
Through this exercise, practical skills were strengthened in data modeling, DAX calculations, dashboard storytelling, and BI publishing workflows.

The key objectives of the assignment were to:

1. Understand the hotel business problem and client reporting needs
2. Load and transform multiple datasets using Power Query
3. Build a star schema data model with correct relationships
4. Create calculated columns and measures using DAX
5. Design an interactive Power BI dashboard for decision-making
6. Publish and share the report on the Power BI Service

Tasks Completed

The assignment began with **loading the hotel datasets** provided via the Codebasics resource into Power BI Desktop. Power Query Editor was used to clean and transform the data by correcting data types, removing duplicates, renaming columns for clarity, and handling missing values where necessary. Dimension tables such as **Date** and **Rooms** were prepared to support time-based and room-level analysis.

Next, a **star schema data model** was built by establishing relationships between fact tables (e.g. bookings or revenue) and dimension tables such as dim_date, dim_rooms, and dim_hotels. This modeling approach improved query performance and ensured accurate aggregations across visuals.

Using **Data Analysis Expressions (DAX)**, several calculated measures were created to support business insights, including metrics such as total revenue, average daily rate (ADR), occupancy rate, booking counts, and revenue trends over time. These measures enabled dynamic and reusable calculations across the report.

An **interactive Power BI dashboard** was then designed to clearly address hotel management decision needs. The dashboard included KPIs, trend analysis, filters, and slicers to allow users to explore performance by date, room type, and hotel category. Visual elements were chosen to clearly communicate insights such as occupancy patterns, revenue contribution, and seasonal trends.

Finally, the completed report was **published to the Power BI Service**, shared publicly, and documented with screenshots as evidence of task completion. The project link was added to my analytics portfolio as part of my professional showcase.

Link to Code:

https://drive.google.com/drive/folders/13BVwERy9uZ6C3Qrit6TAKLmmQcOb5TGO?usp=sharing

Conclusion

This assignment provided practical exposure to **end-to-end Business Intelligence development** using Power BI within a hotel management context. By transforming raw operational data into a structured data model and meaningful dashboards, the project demonstrated how BI tools support strategic and operational decisions in the hospitality industry.

The exercise strengthened skills in data transformation, dimensional modeling, DAX, and dashboard design, all of which are essential for modern data analyst and BI roles. The completed Power BI report and documentation serve as a valuable addition to my professional data analytics portfolio.

Screenshots of the project

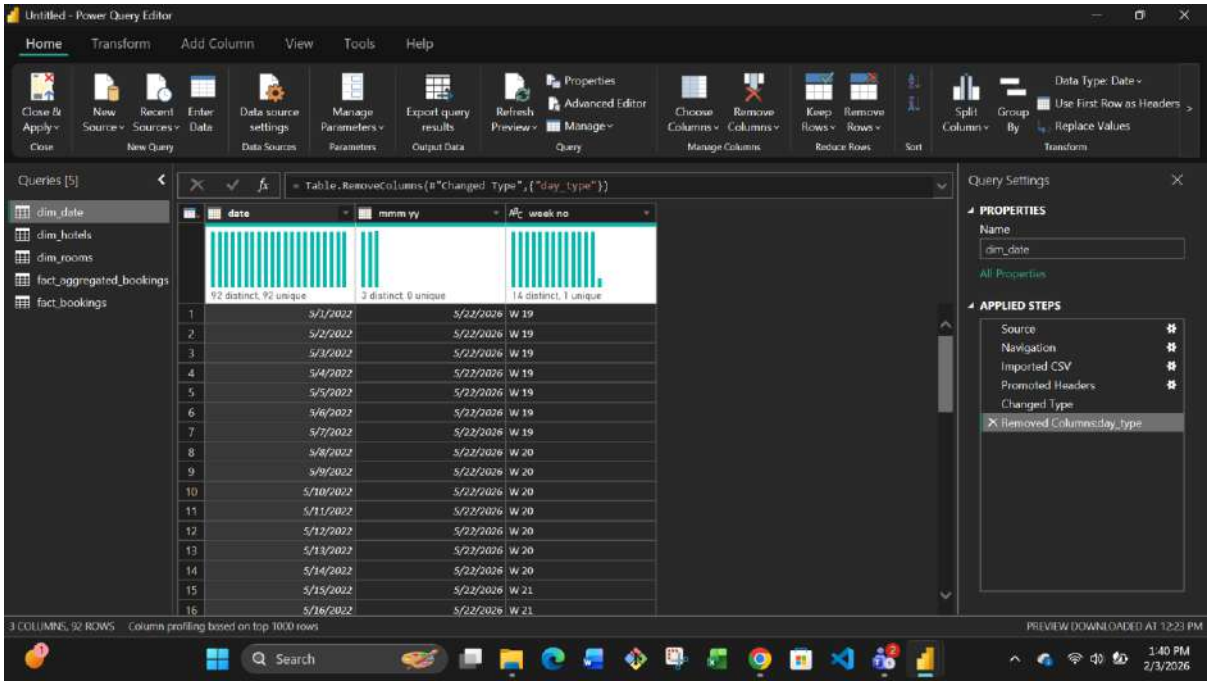
The screenshots show the key stages of the **Power BI Hotel Management analysis**, including data loading and transformation in Power Query, star schema data modelling, DAX measure creation, and the final interactive dashboard. These steps demonstrate how raw hotel data was transformed into meaningful insights to support data-driven decision-making.

Task 1: Power Query data transformation steps

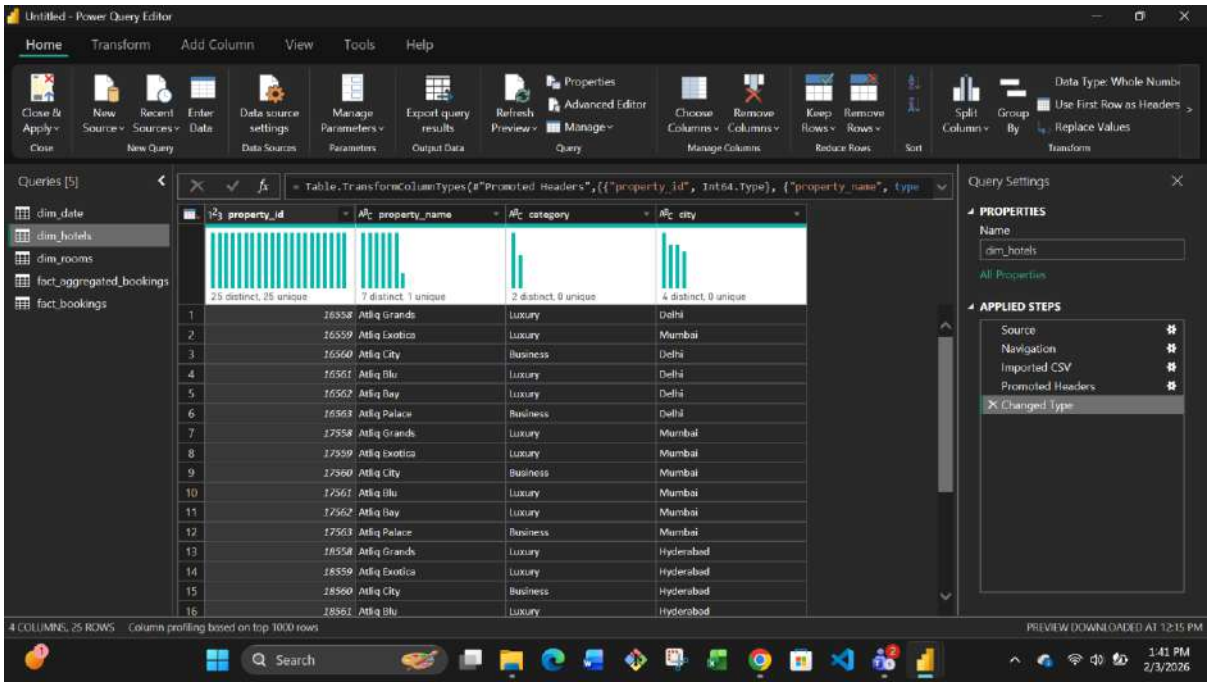
Preparation and cleaning of raw hotel data before analysis.
What:
Data was cleaned, data types were corrected, columns were renamed, duplicates were removed, and missing values were handled using Power Query.
Why:
Clean and well-structured data is essential for accurate analysis and reliable reporting.

LOAD, TRANSFORM FOR EACH OF THE QUERIES

1. **dim_date**
   Date dimension used for time-based analysis and trend reporting.
   Load: The date dataset was loaded into Power BI from the provided source.
   Transform: Date columns were formatted correctly, additional date attributes (year, month, quarter) were created, and duplicates were removed.
   Close & Apply: The cleaned date table was applied to enable accurate time intelligence across the dashboard.



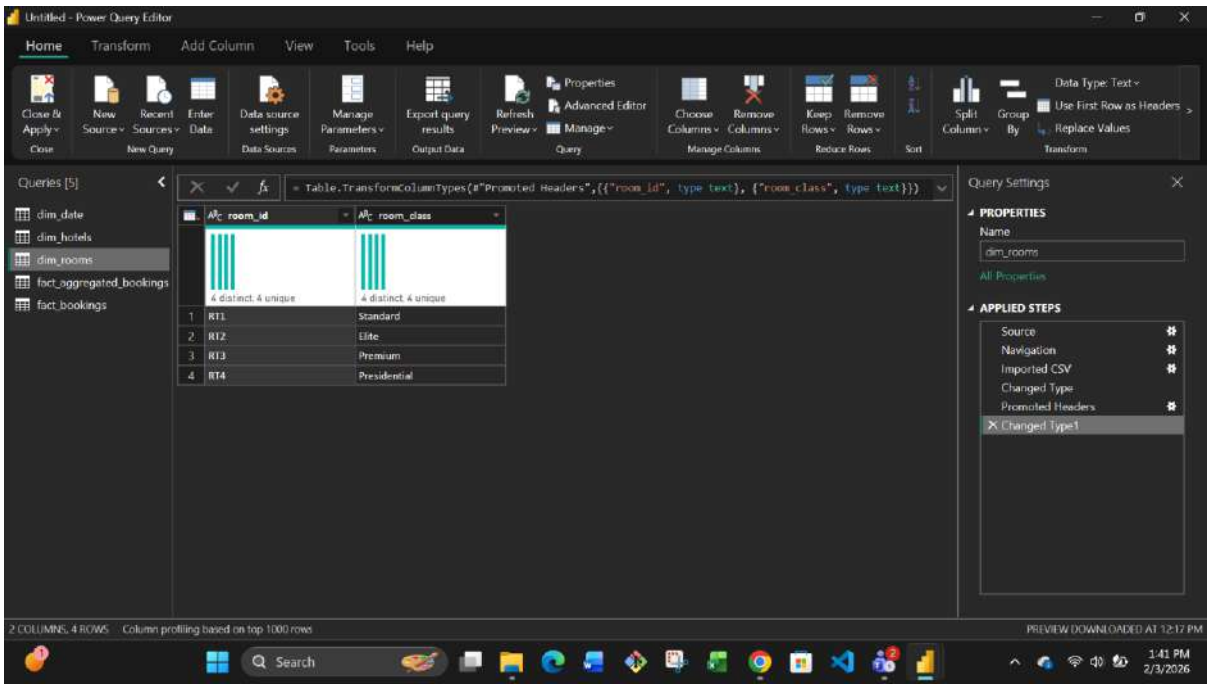
2. dim_hotels
   Dimension table containing hotel-level details.
   Load: Hotel information was imported into Power BI.
   Transform: Column names were standardized, data types were corrected, and irrelevant fields were removed.
   Close & Apply: The refined hotel dimension was applied for consistent hotel-level filtering and analysis.



3. dim_rooms
   Dimension table describing room categories and attributes.
   Load: Room-related data was loaded into Power BI.
   Transform: Room types were cleaned, duplicates were removed, and categorical fields were

standardized.

Close & Apply: The prepared room dimension was applied to support room-level performance analysis.



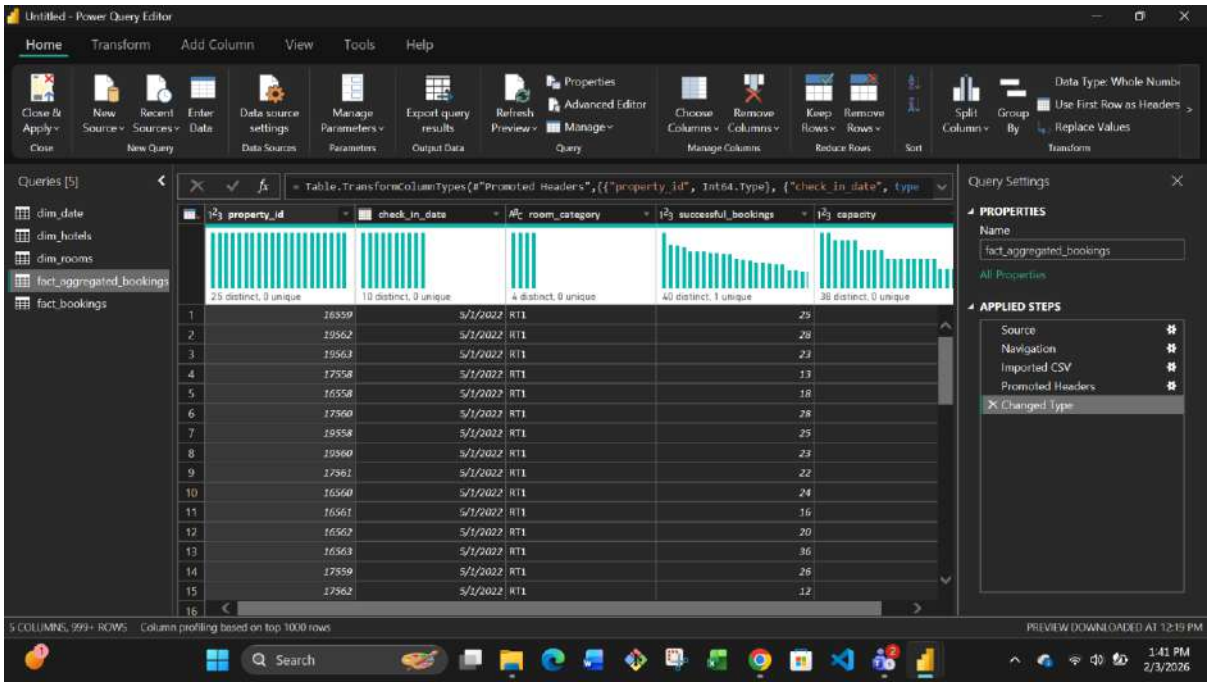
4. fact_aggregated_bookings

   Aggregated fact table summarizing booking performance metrics.

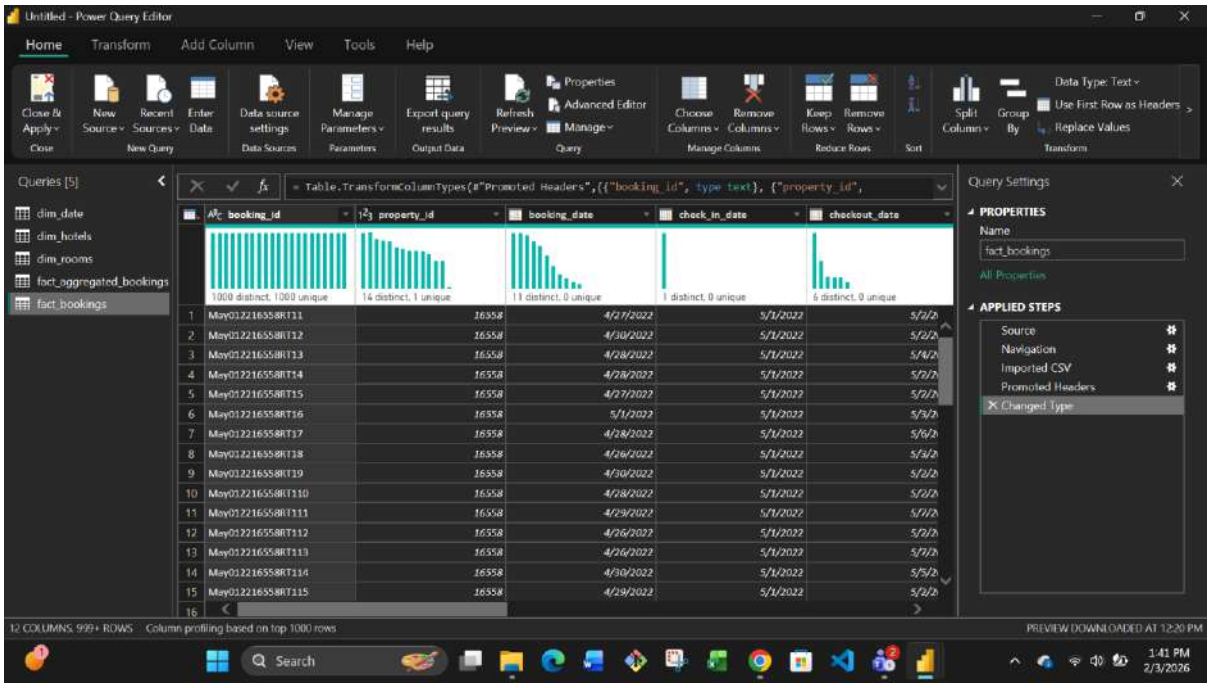
   Load: Aggregated booking data was loaded into Power BI.

   Transform: Metrics were validated, aggregation levels were confirmed, and data types were corrected.

   Close & Apply: The finalized aggregated data was applied to enable high-level KPI reporting and trend analysis.



5. fact_bookings
   Fact table capturing individual hotel booking transactions.
   Load: Booking transaction data was imported into Power BI.
   Transform: Date fields were standardized, numerical columns were validated, and unnecessary columns were removed.
   Close & Apply: The transformed fact table was applied to support detailed booking and revenue analysis.

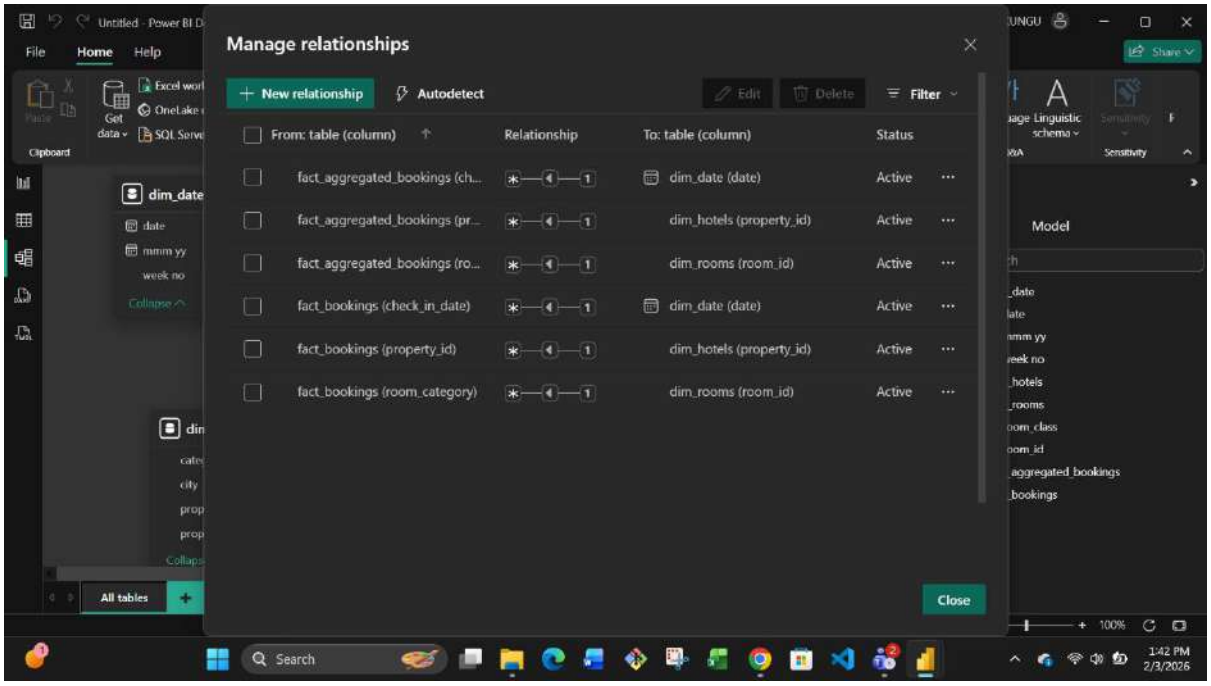


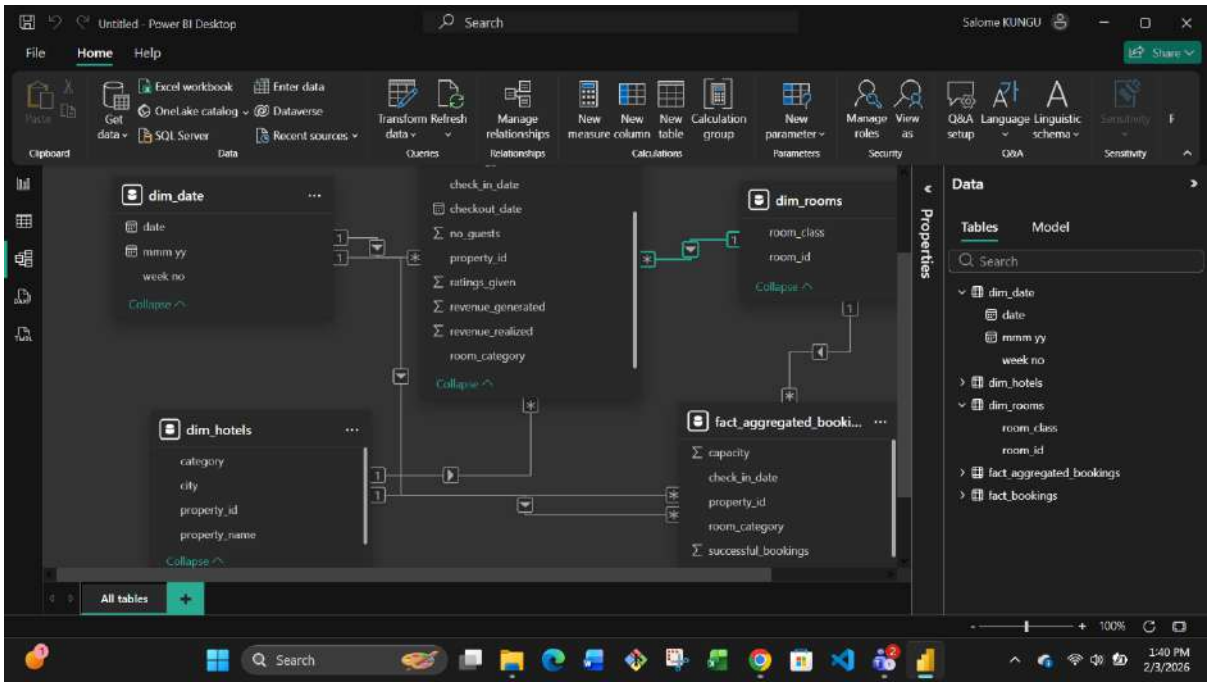
Task 2: Star Schema Data Model Relationships

Structuring the data model for efficient analysis. Star schema is for building the underlying data model and the required relationships.

What: Fact tables were linked to dimension tables such as date and rooms using a star schema design.

Why: This improves performance and ensures correct aggregations across all visuals.





Task 3: DAX Measures Used for KPIs

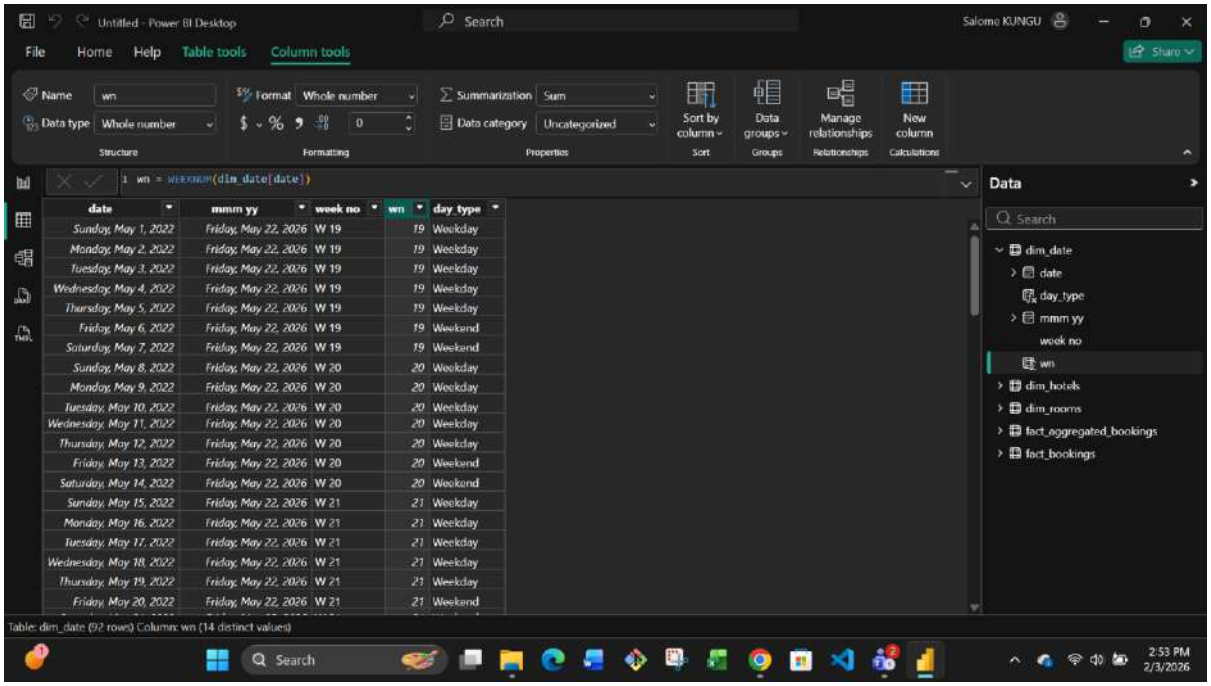
Creation of calculated metrics for business insights.

What: DAX measures were created to calculate key metrics such as total revenue, occupancy rate, and average daily rate (ADR).

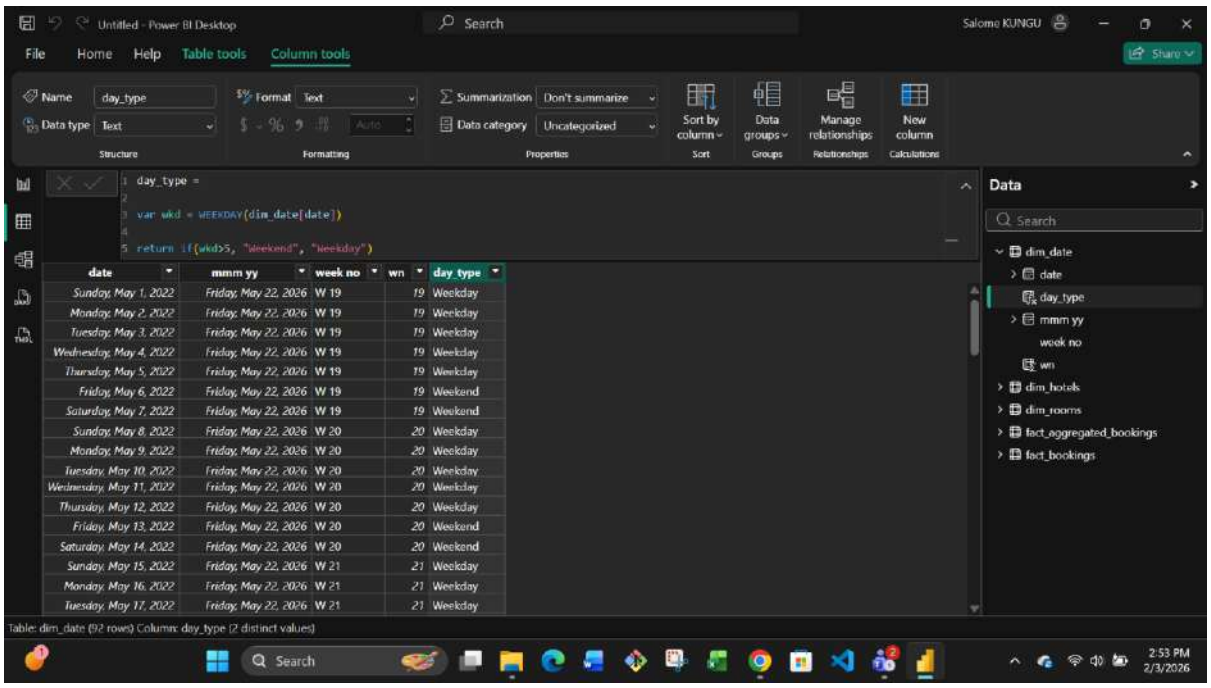
Why: These measures enable dynamic calculations and provide meaningful business insights for decision-making. DAX allows dynamic calculations that update automatically based on filters and slicers.

DAX MEASURES AND CALCULATE WITH BRIEF DESCRIPTIONS

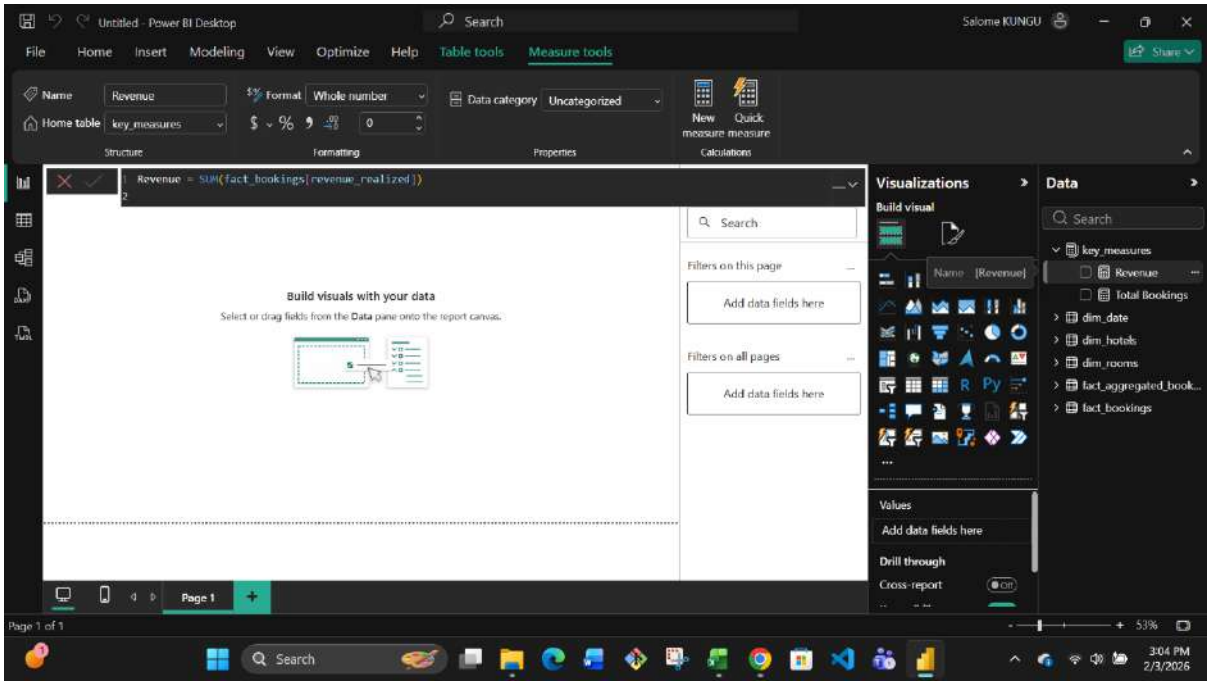
1. Calculate weeknumber(wn). To get the week number from the corresponding date.



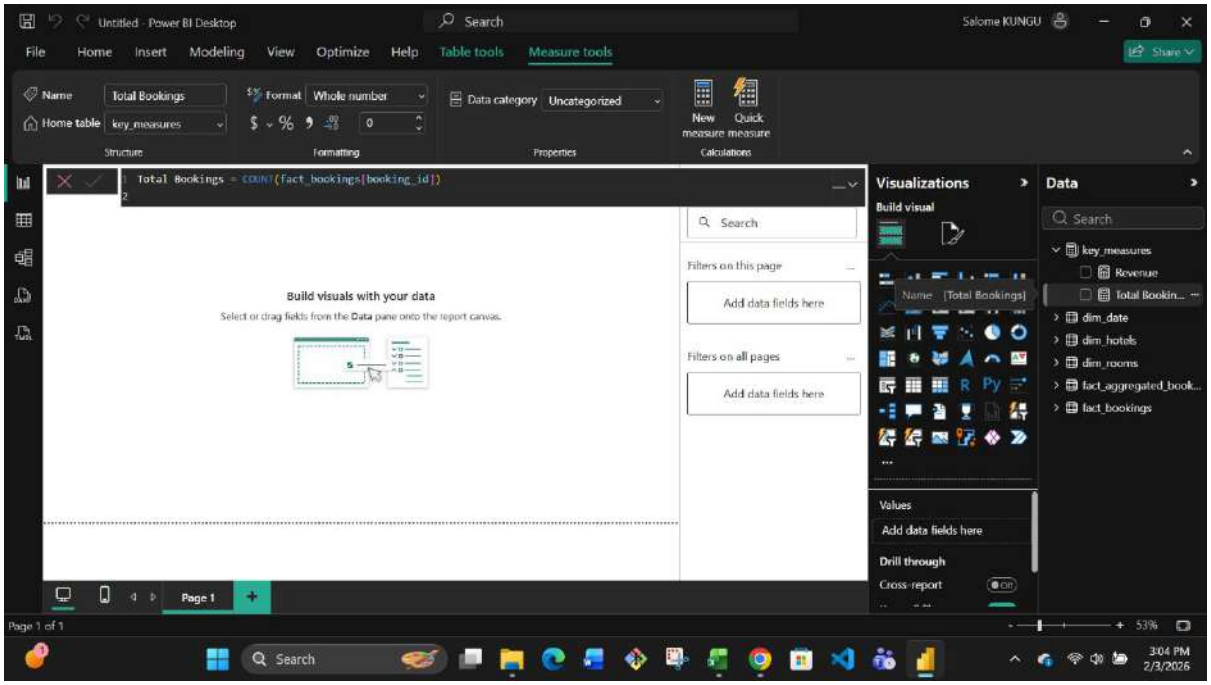
2. Calculate day type.
   Based on the feedback from stakeholder, we considered Friday and Saturday as weekend and weekdays from Sunday to Thurdsay. In PowerBI, Sunday weekday number is 1, Monday is 2 and so on. So, if weekday number is greater than 5, then weekend or else weekday.



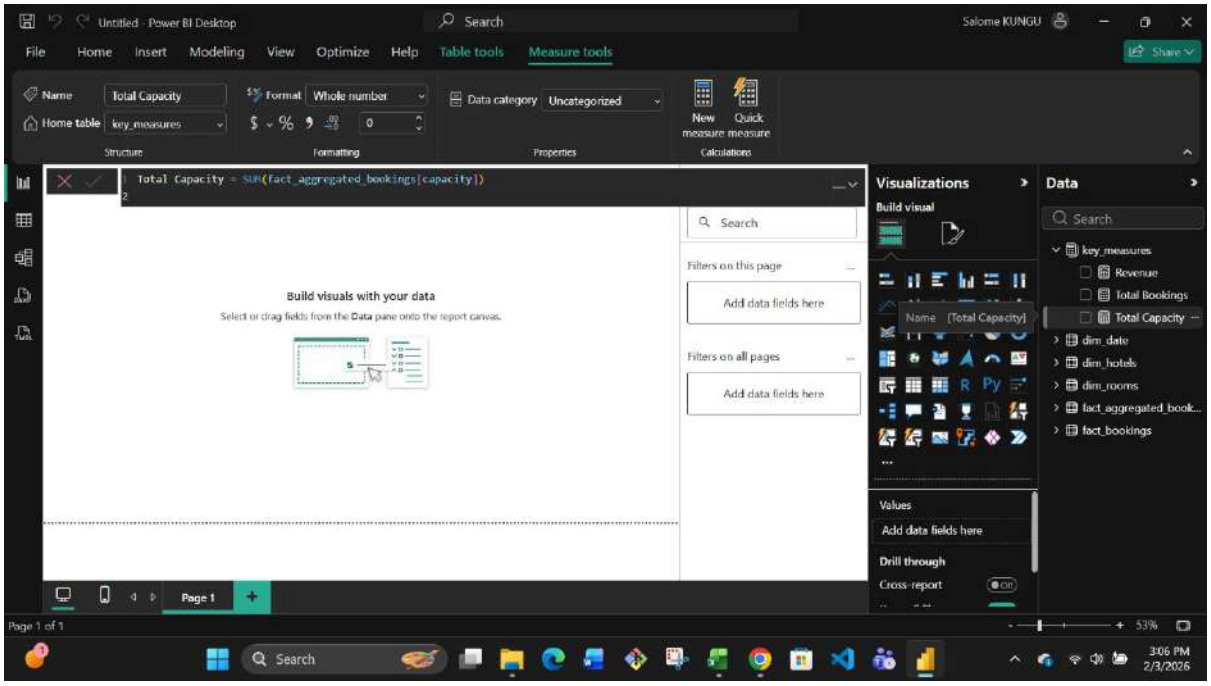
3. Calculate Revenue. To get the total revenue_realized.



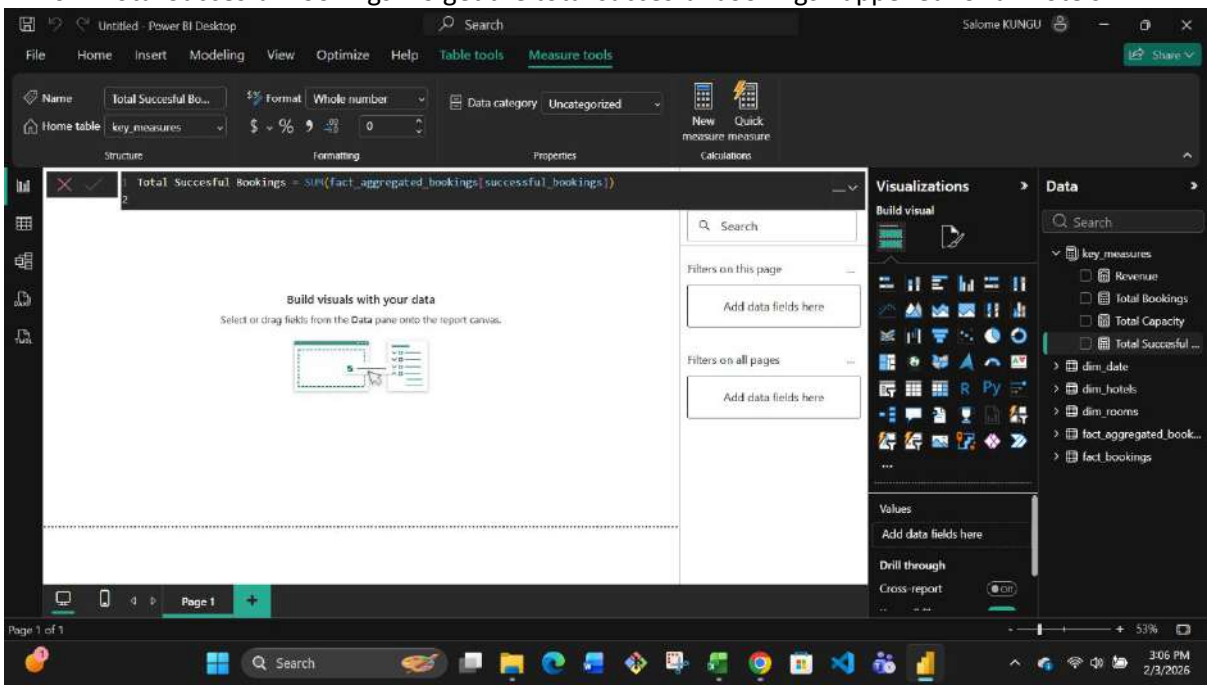
4. Calculate Total Bookings. To get the total number of bookings happened.



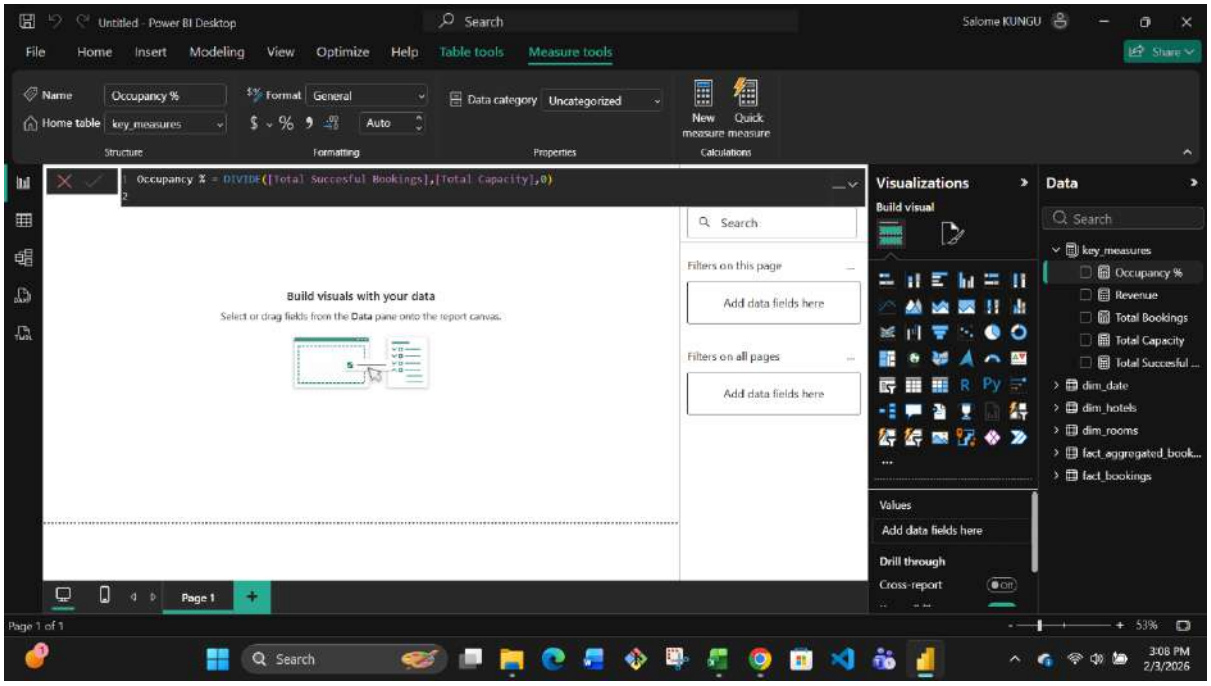
5. Calculate Total Capacity. To get the total capacity of rooms present in hotels.



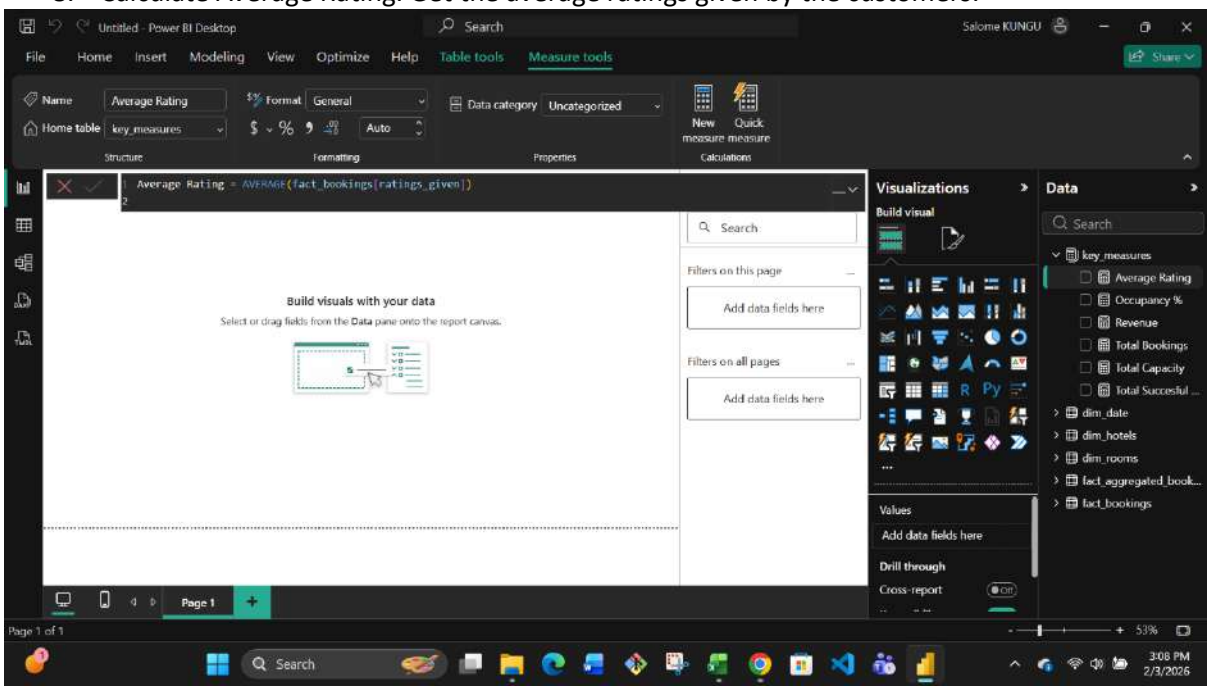
6. Total Succesful Bookings. To get the total succesful bookings happened for all hotels.



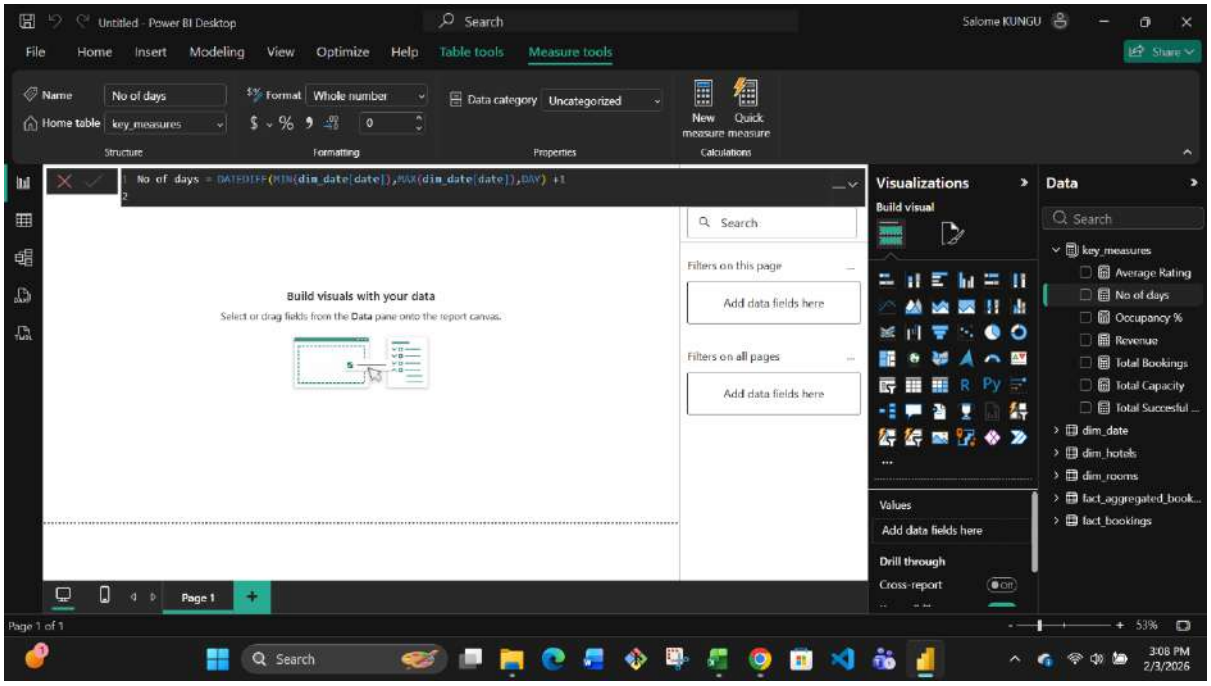
7. Calculate Occupancy %. Occupancy means total successful bookings happened to the total rooms available(capacity)



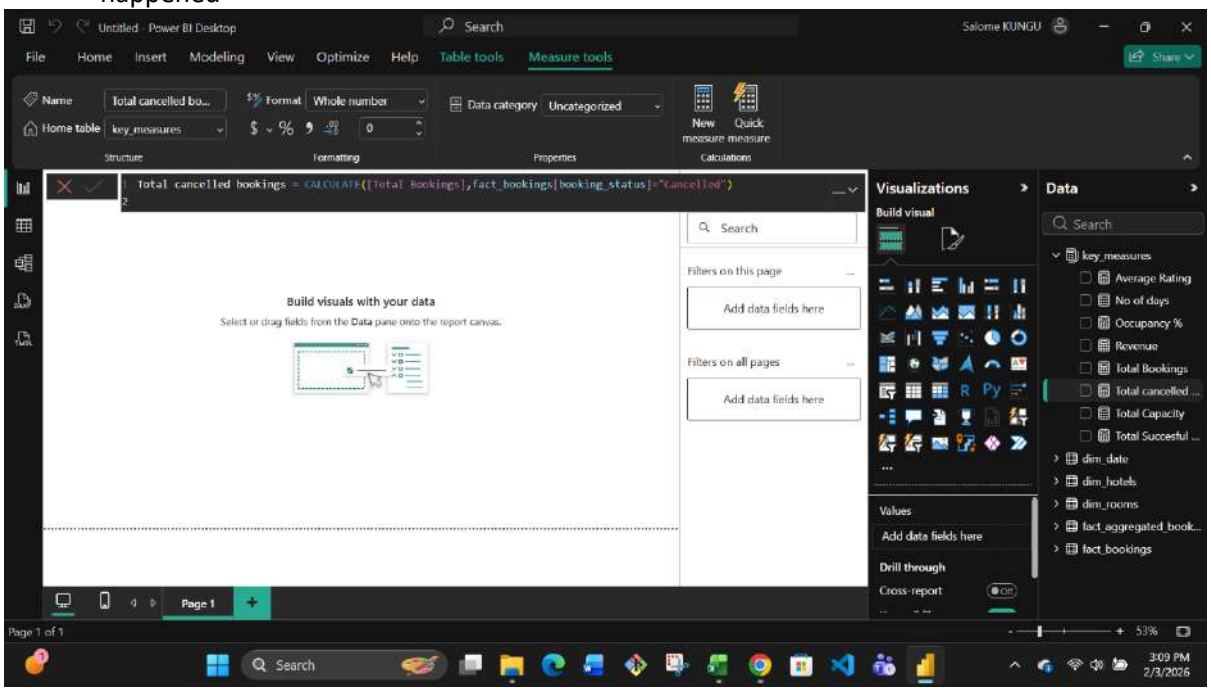
8. Calculate Average Rating. Get the average ratings given by the customers.



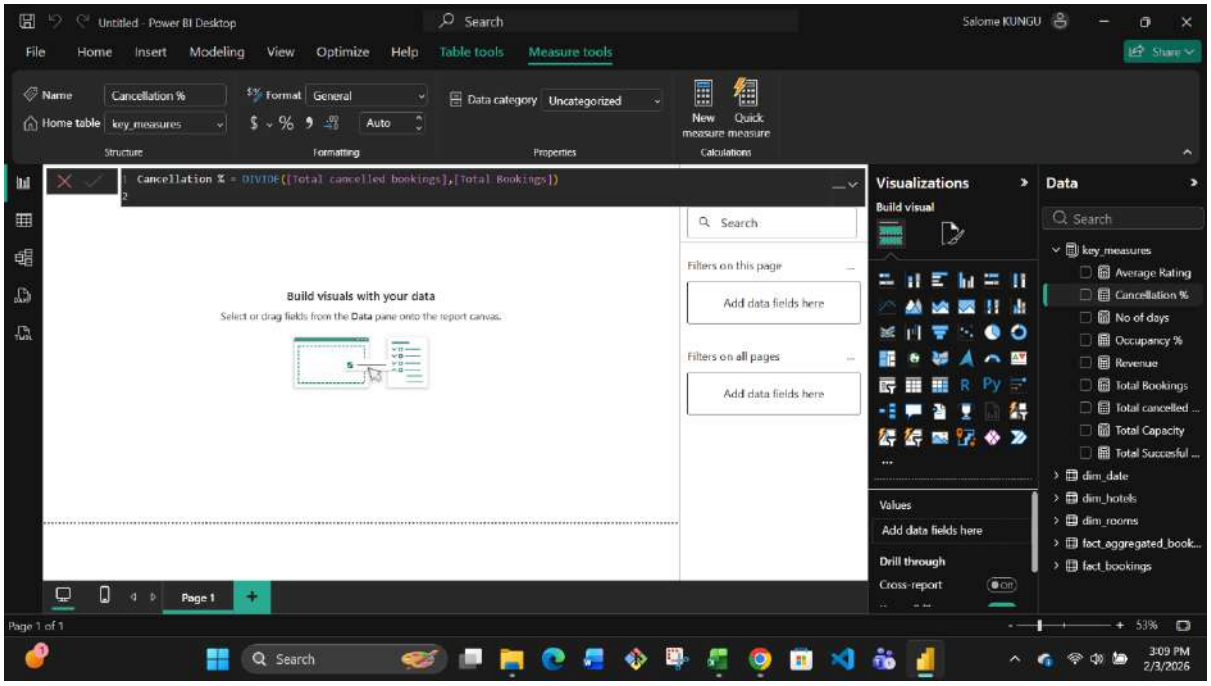
9. Calculate No of days. To get the total number of days present in the data. In our case, we have data from May to July. So 92 days.



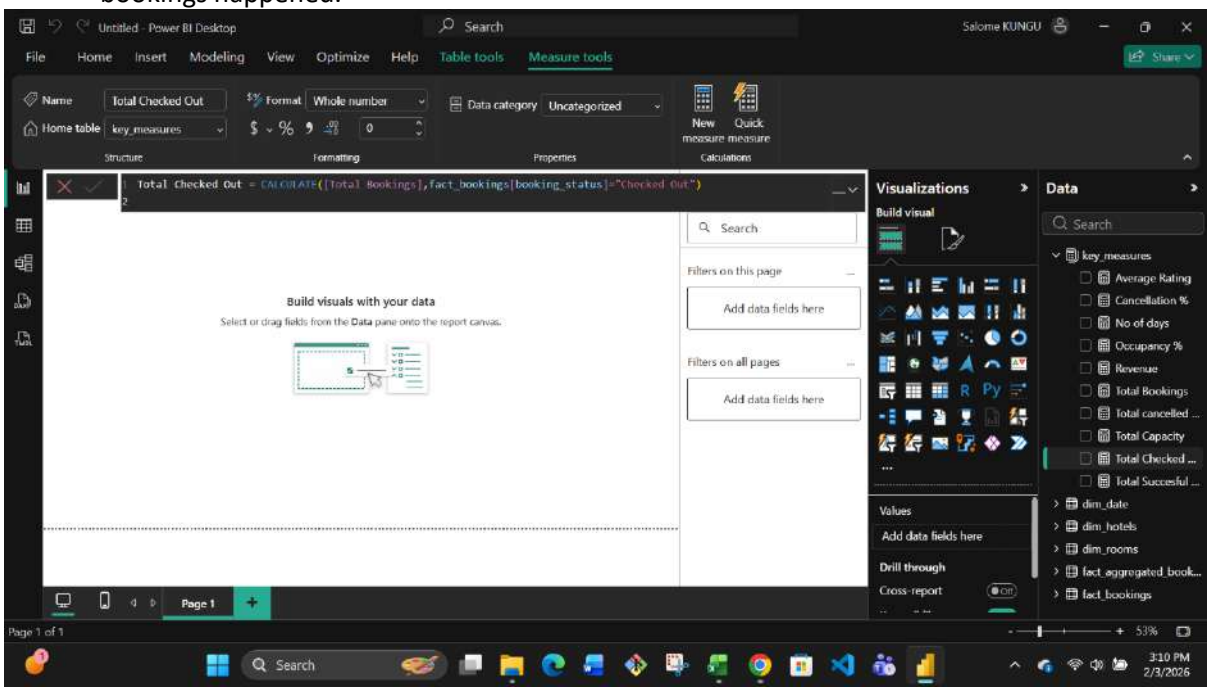
10. Calculate Total cancelled bookings. To get the"Cancelled" bookings out of all Total bookings happened



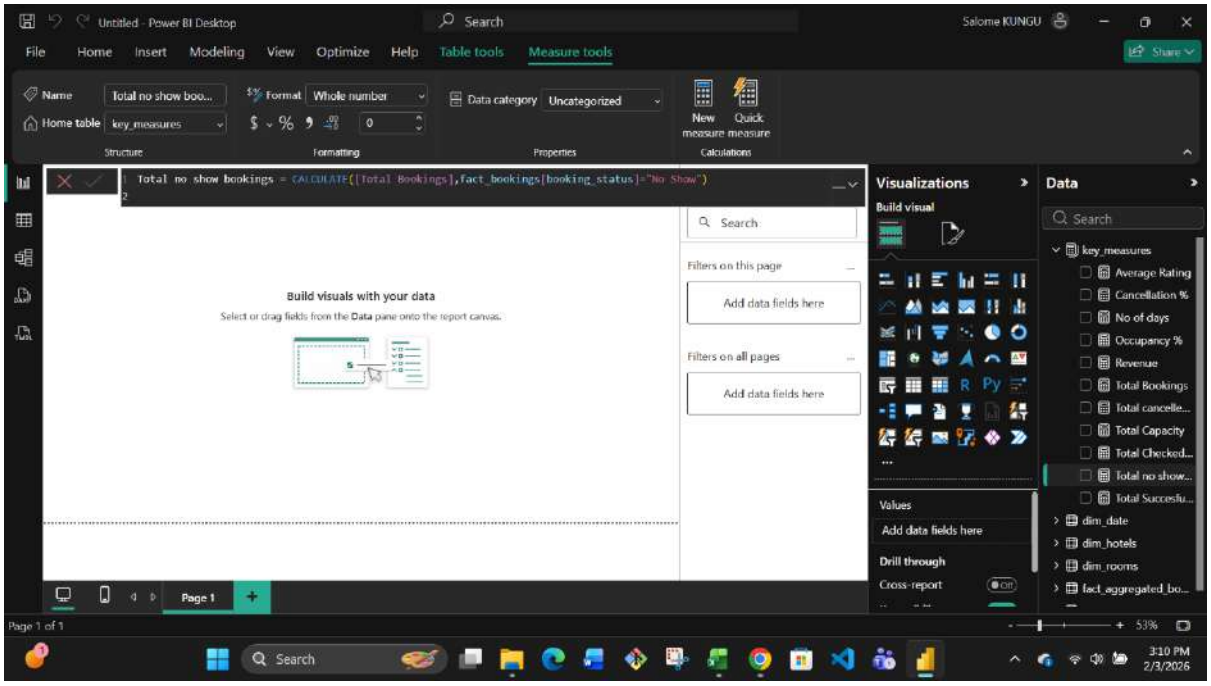
11. Calculate Cancellation %. Calculating the cancellation percentage.



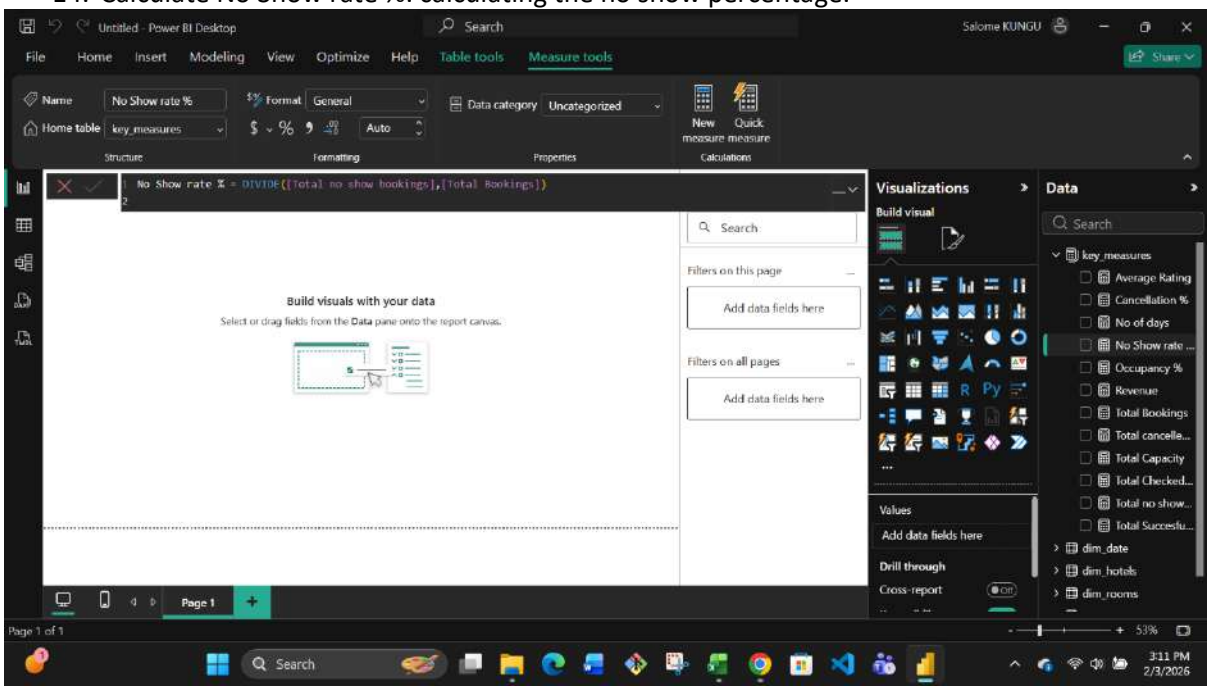
12. Calculation Total Checked Out. To get the successful 'Checked out' bookings out of all Total bookings happened.



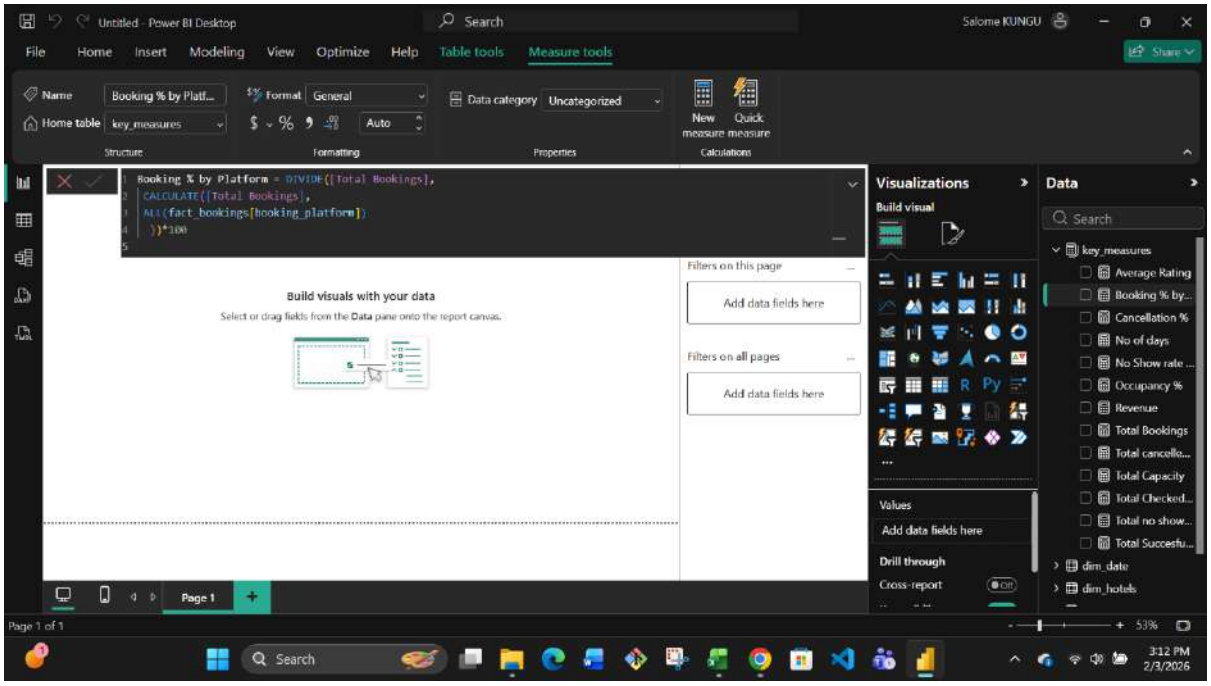
13. Calculate Total no show bookings. To get the"No Show" bookings out of all Total bookings happened. ("No show" means those customers who neither cancelled nor attend to their booked rooms).



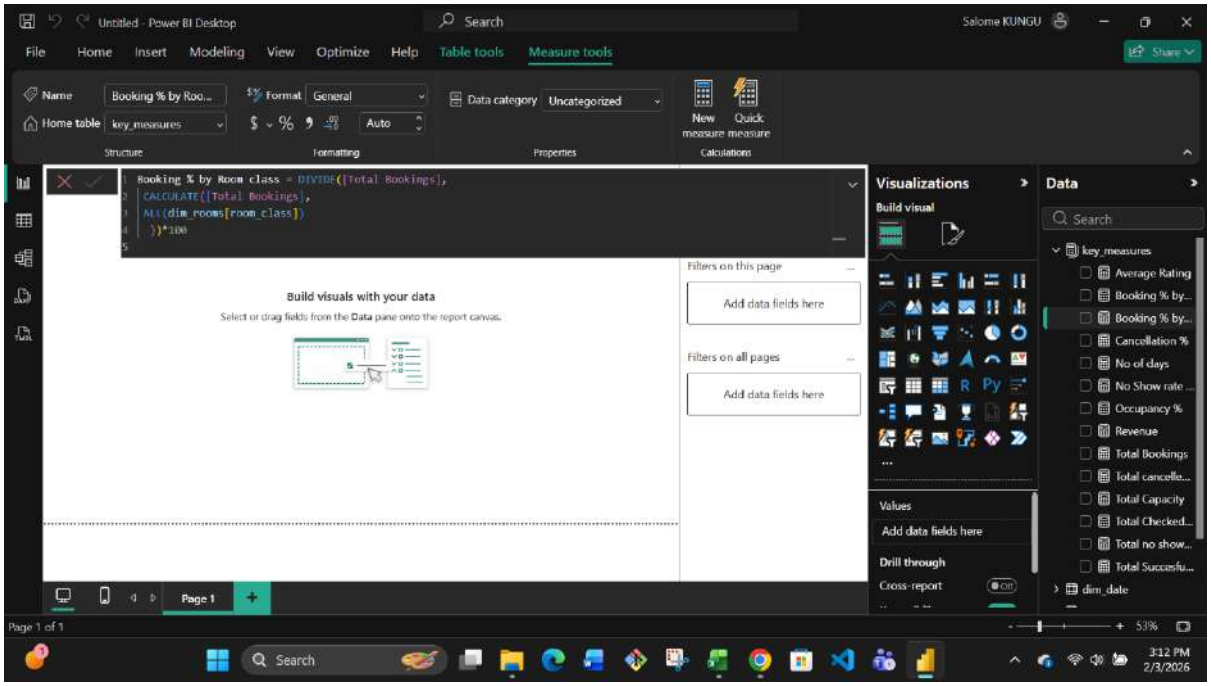
14. Calculate No Show rate %. calculating the no show percentage.



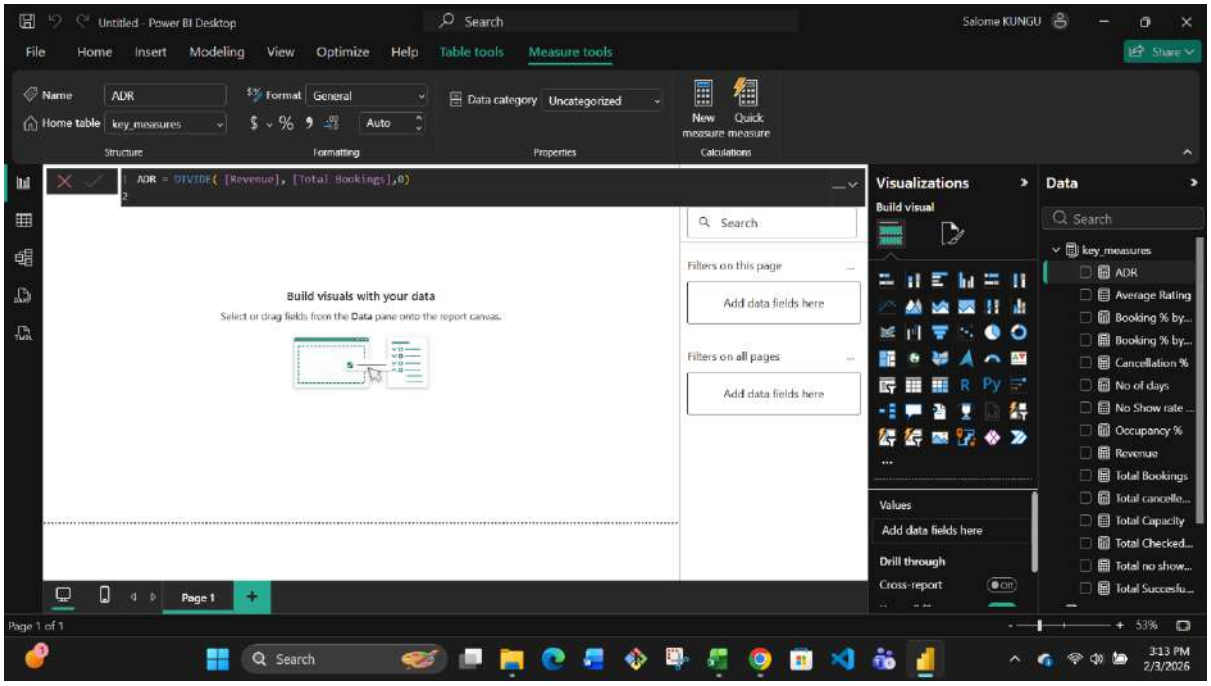
15. Calculate Booking % by Platform. To show the percentage contribution of each booking platform for bookings in hotels.



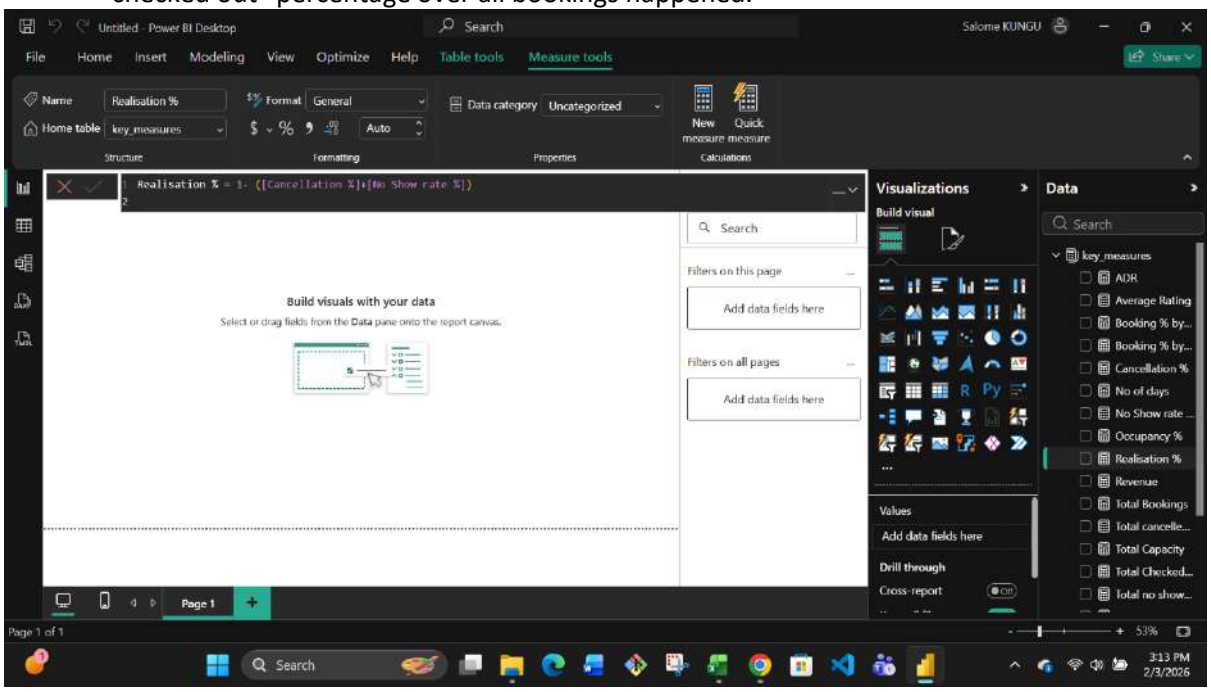
16. Calculate Booking % by Room class. To show the percentage contribution of each room class over total rooms booked.



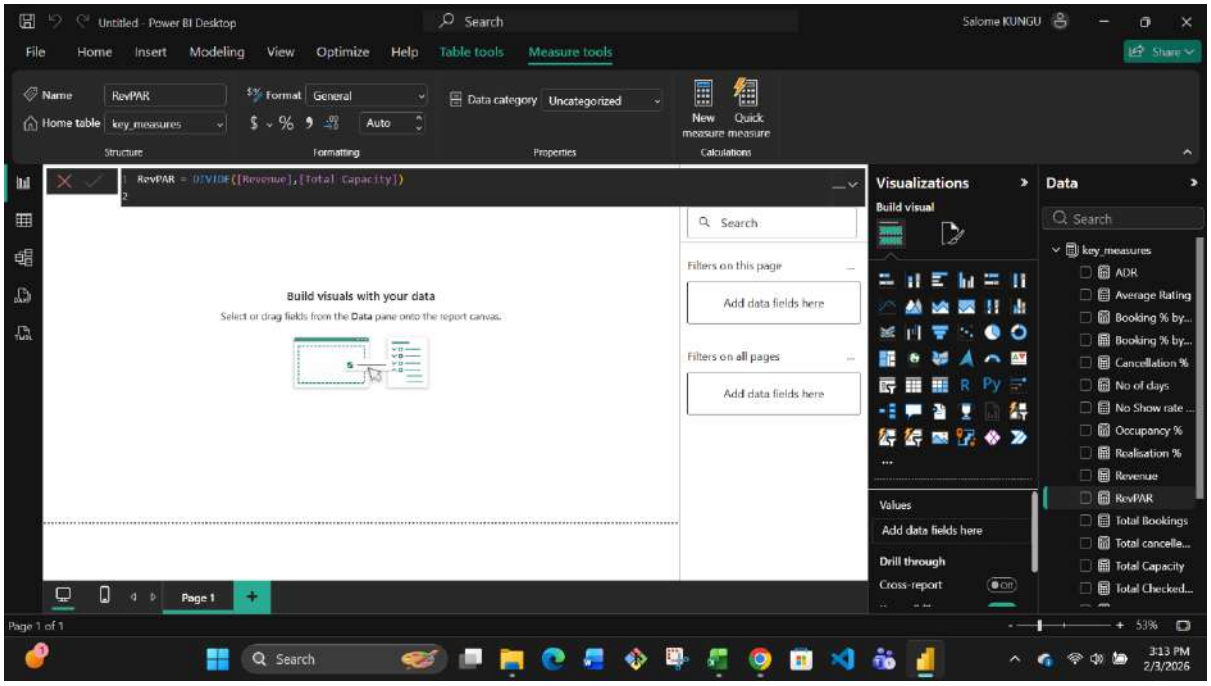
17. Calculate ADR. Calculate the ADR(Average Daily rate). It is the ratio of revenue to the total rooms booked/sold. It is the measure of the average paid for rooms sold in a given time period.



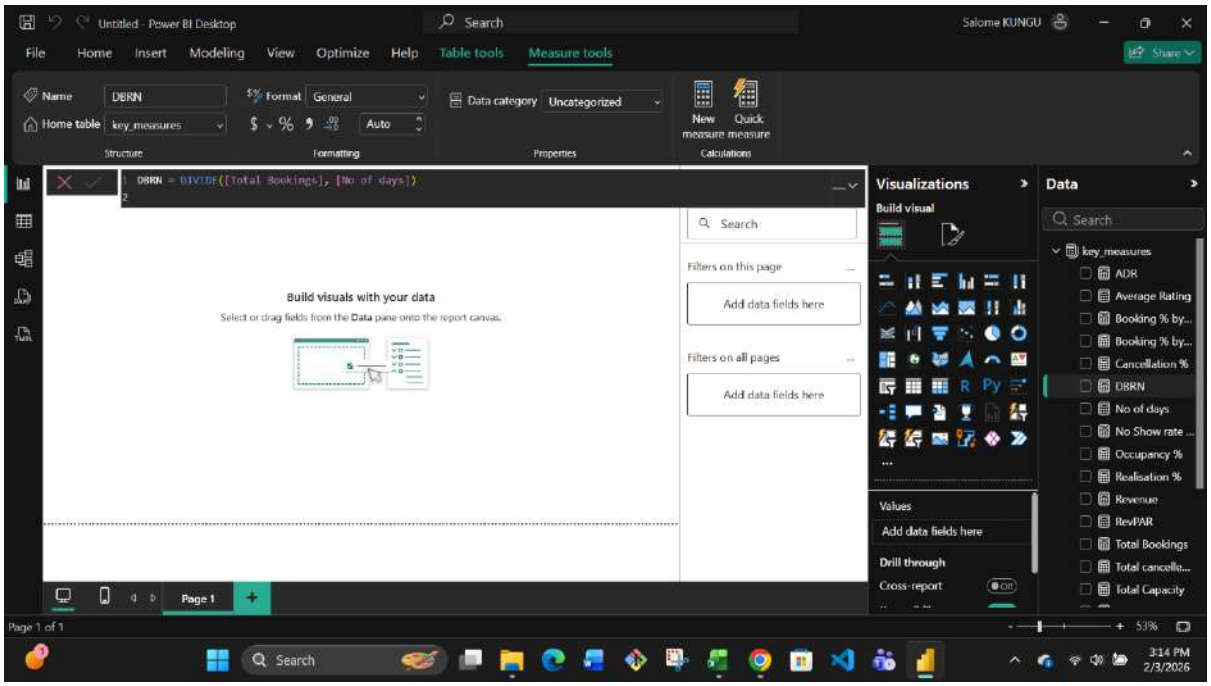
18. Calculate Realisation %. Calculate the realisation percentage. It is nothing but the succesful "checked out" percentage over all bookings happened.



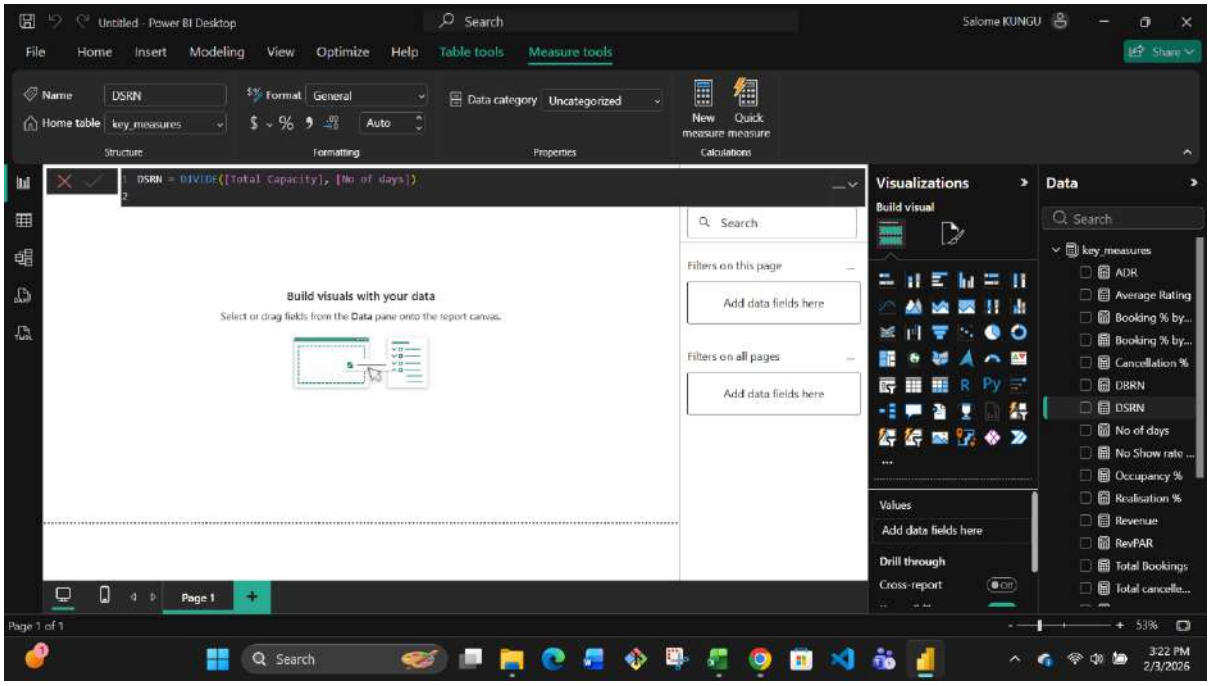
19. Calculate RevPAR. Calculate the RevPAR (Revenue Per Available Room). RevPAR represents the revenue generated per available room, whether or not they are occupied. RevPAR helps hotels measure their revenue generating performance to accurately price rooms. RevPAR can help hotels measure themselves against other properties or brands.



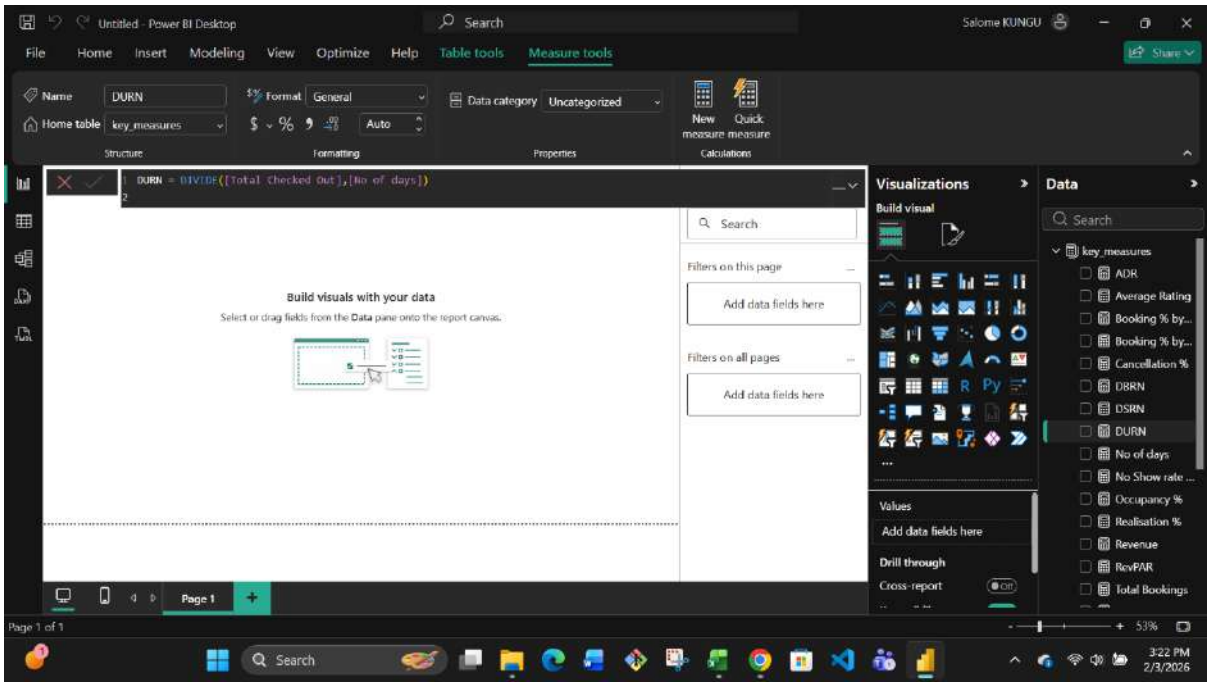
20. Calculate DBRN. calculate DBRN (Daily Booked Room Nights). This metrics tells on average how many rooms are booked for a day considering a time period.



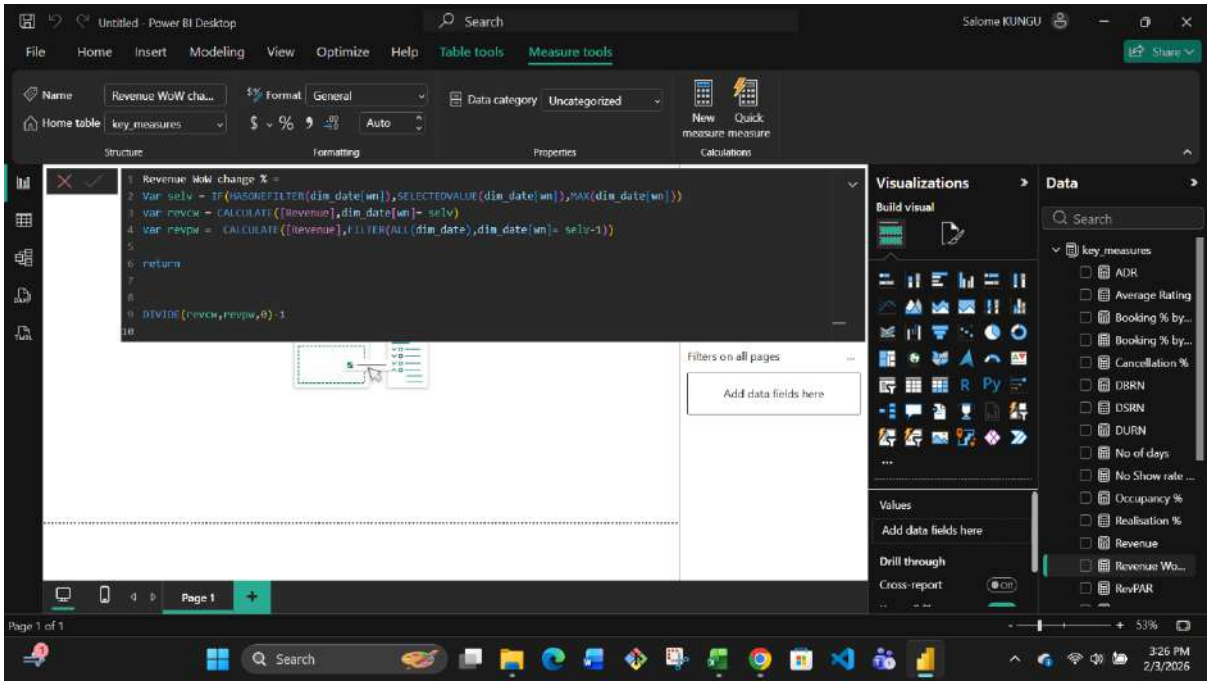
21. Calculate DSRN. Calculate DSRN (Daily Sellable Room Nights). This metrics tells on average how many rooms are ready to sell for a day considering a time period



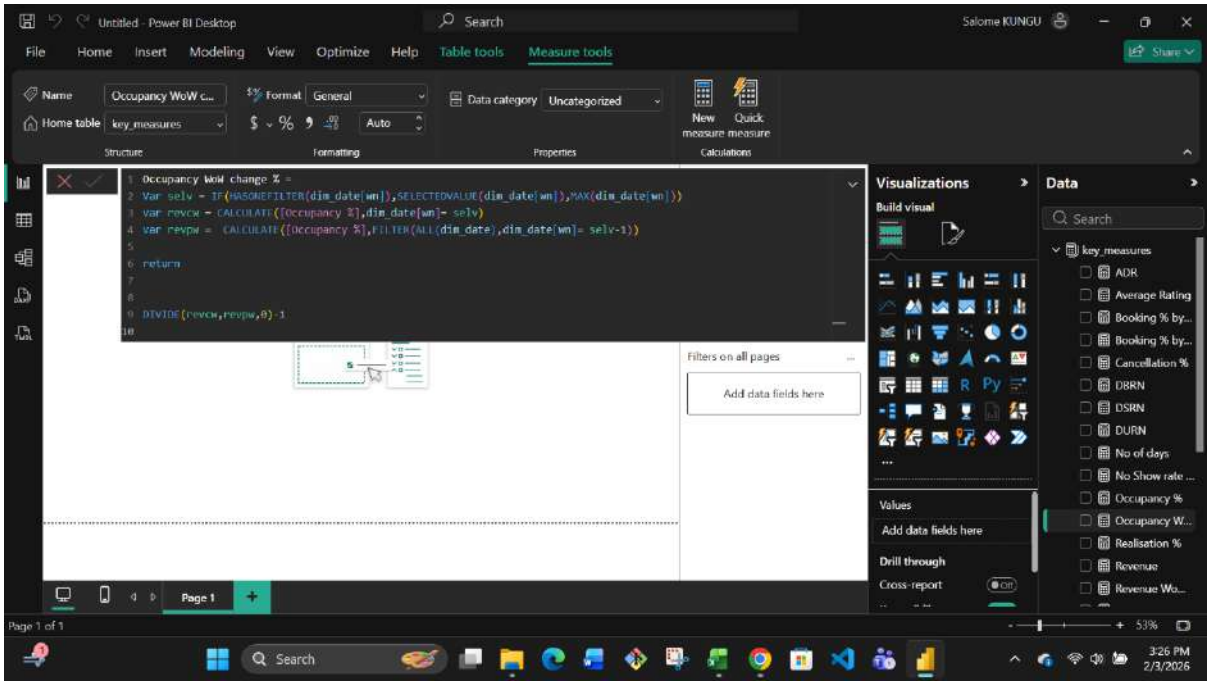
22. Calculate DURN. Calculate DURN (Daily Utilized Room Nights). This metric tells on average how many rooms are succesfully utilized by customers for a day considering a time period



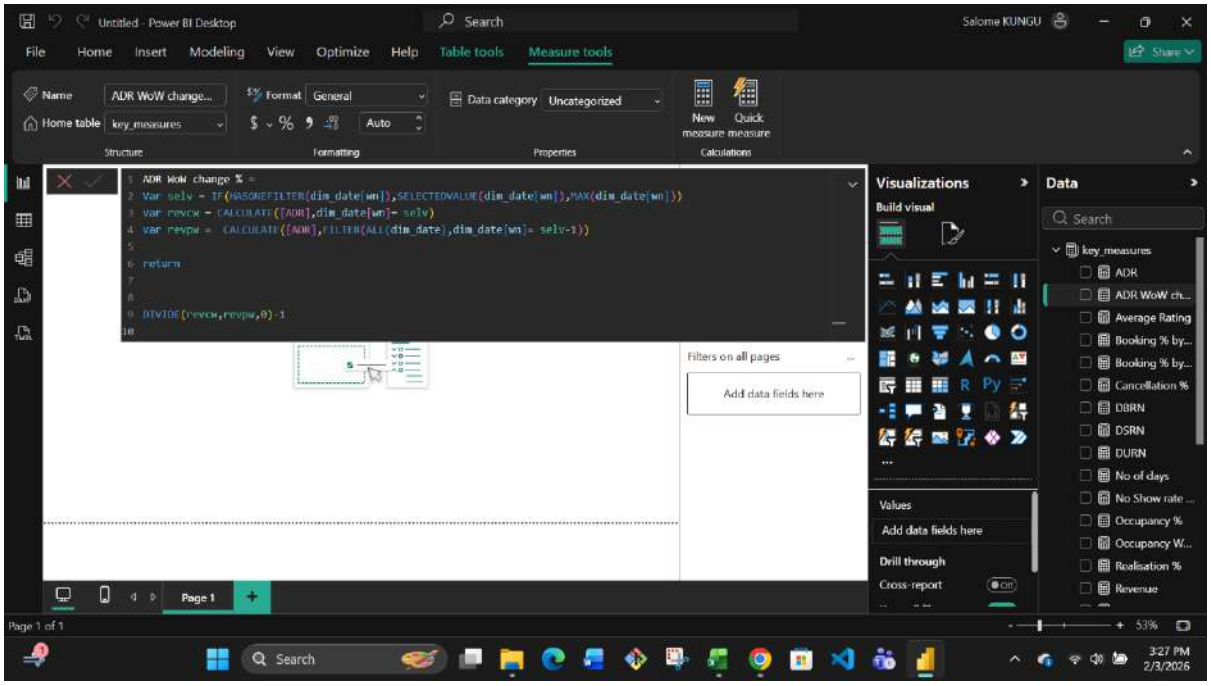
23. Revenue WoW change %. To get the revenue change percentage week over week.



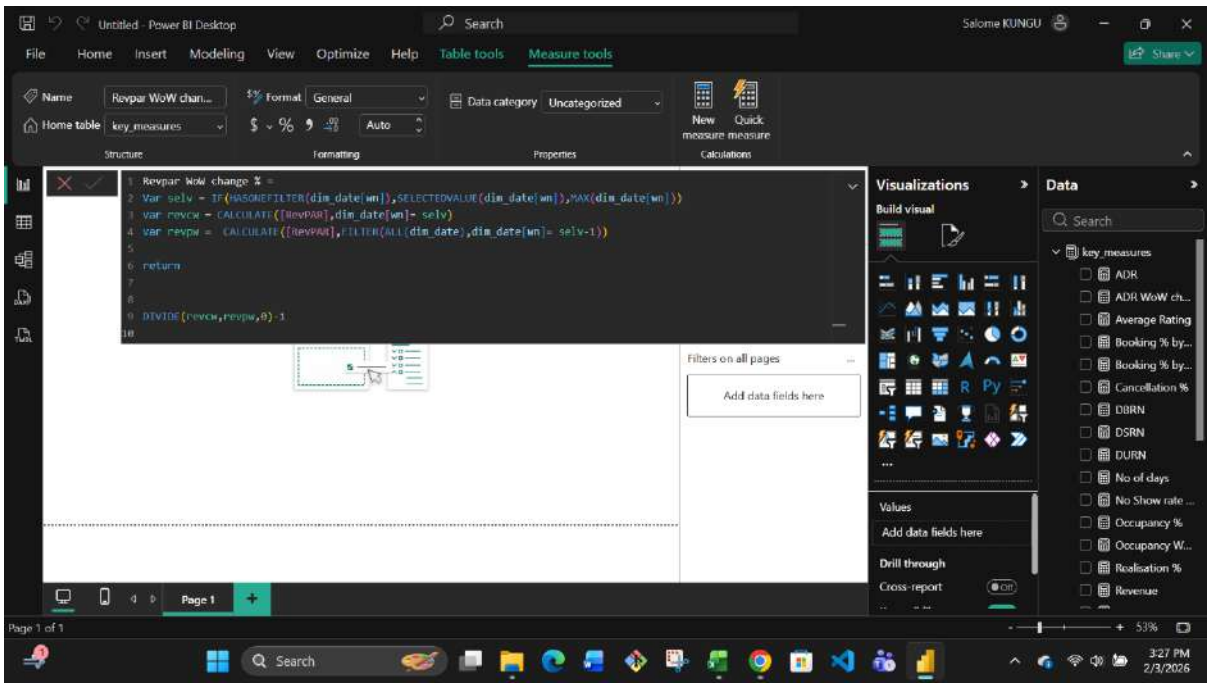
24. Calculate Occupancy WoW change %. To get the occupancy change percentage week over week.



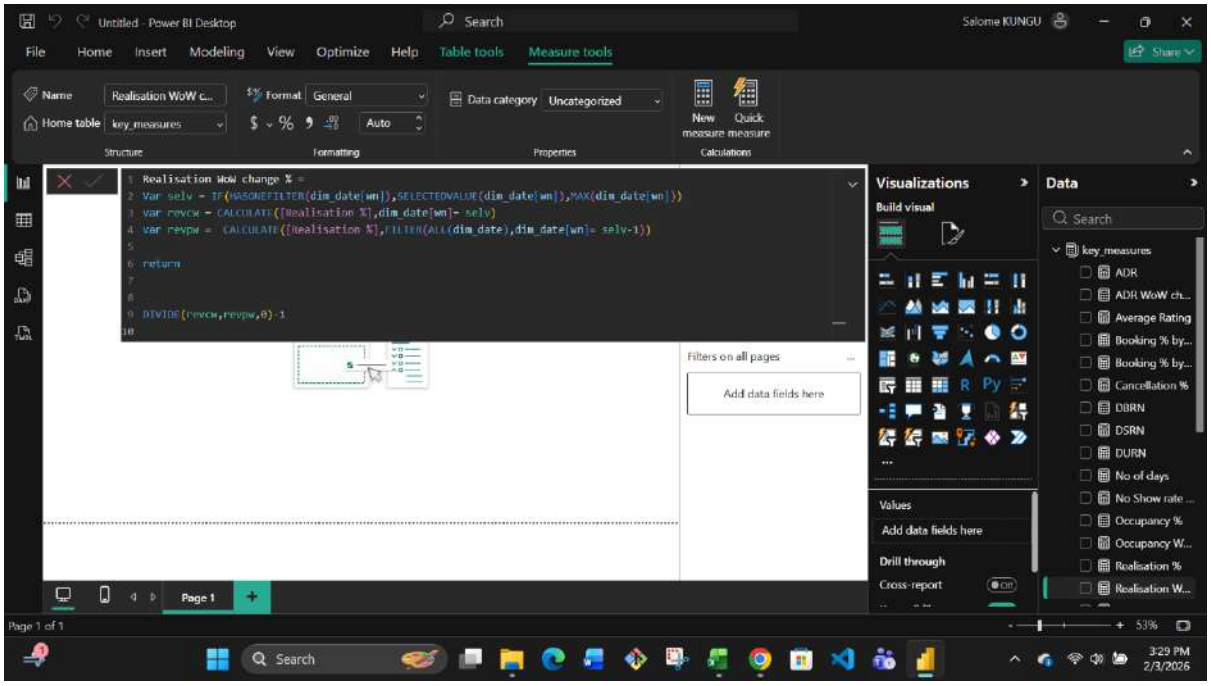
25. Calculate ADR WoW change %. To get the ADR(Average Daily rate) change percentage week over week.



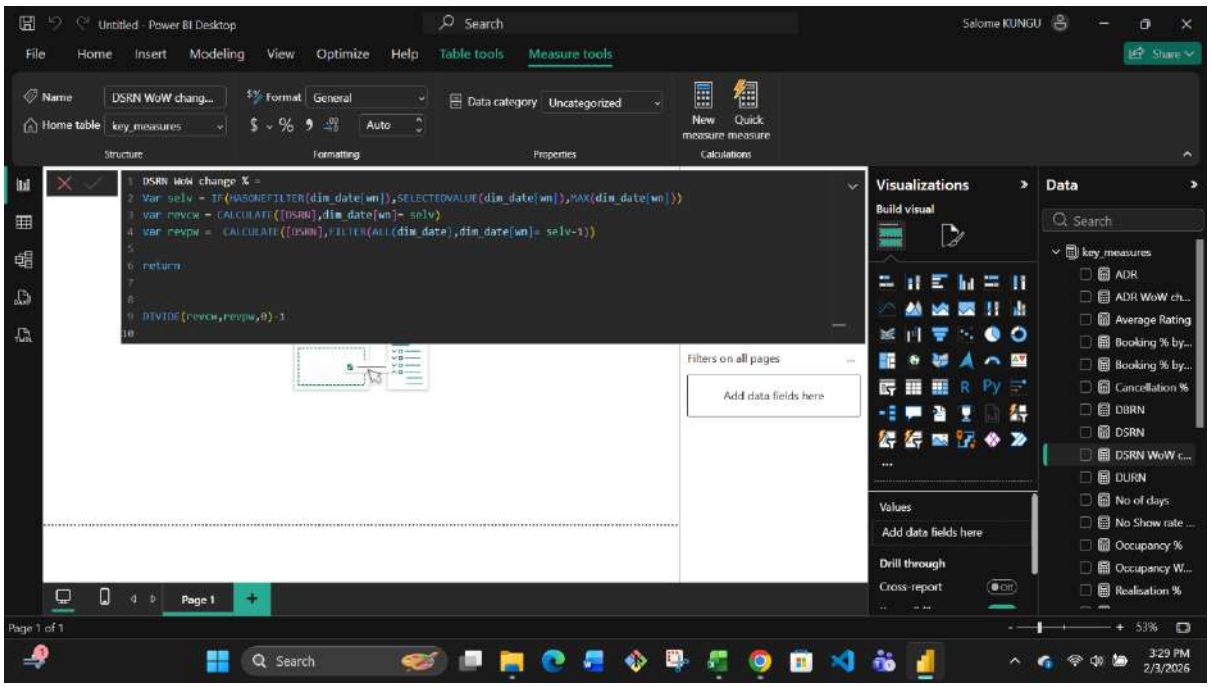
26. Calculate Revpar WoW change %. To get the RevPar(Revenue Per Available Room) change percentage week over week.



27. Calculate Realisation WoW change %. To get the Realisation change percentage week over week.



28. Calculate DSRN WoW change %. To get the DSRN(Daily Sellable Room Nights) change percentage week over week.

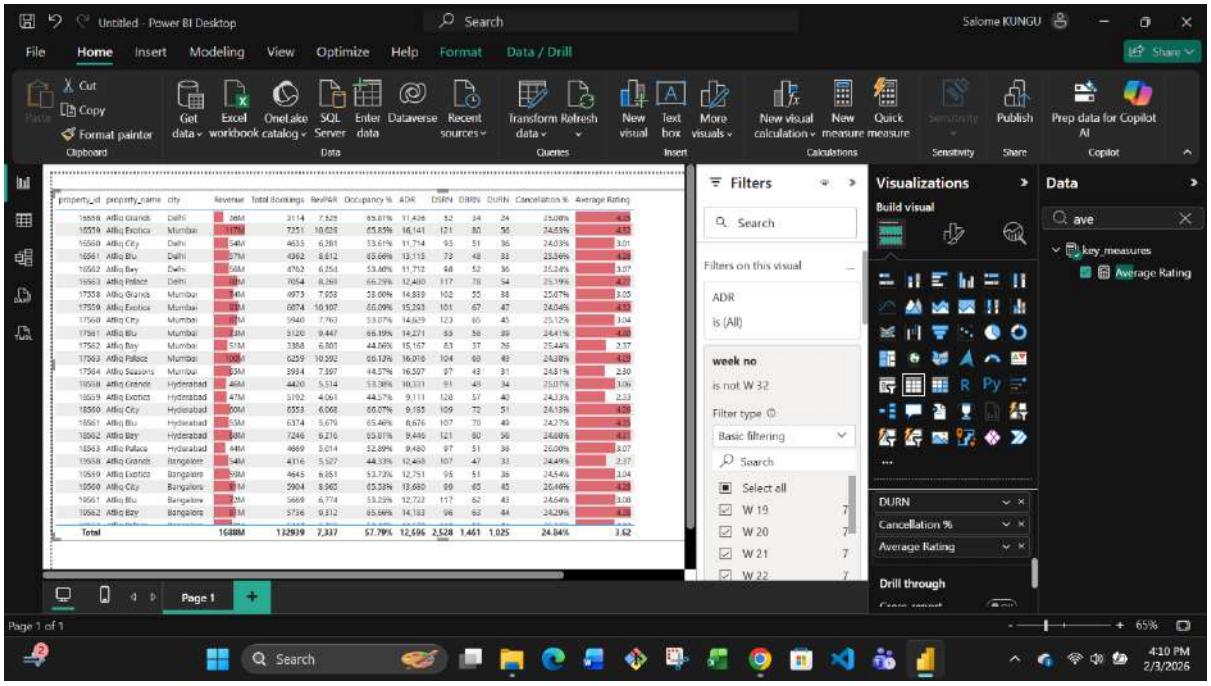


Task 4: Interactive Dashboard Visuals and Slicers

Visual presentation of insights with user interactivity.

What: Charts, KPI cards, and slicers were used to visualize hotel performance across different dimensions.

Why: Interactivity enables stakeholders to easily explore trends and make informed decisions.


Untitled - Power BI Desktop
Search
Salome KUNGU
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Publish
Share
Prep data for Copilot AI
Copilot
Filters
Search
Filters on this visual
ADR
is (All)
week no
is not W 32
Filter type
Basic filtering
Select all
W 19
W 20
W 21
W 22
Visualizations
Build visual
DURN
Cancellation %
Average Rating
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Data
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key_measures
Average Rating
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Page 1 of 1
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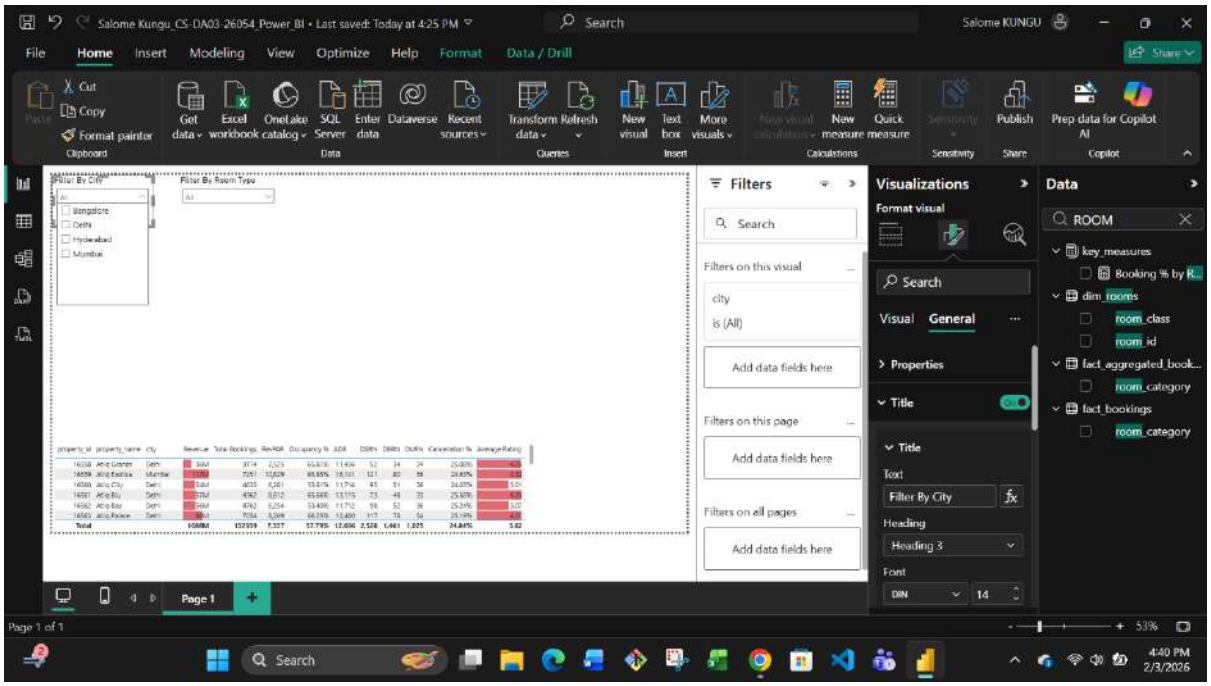
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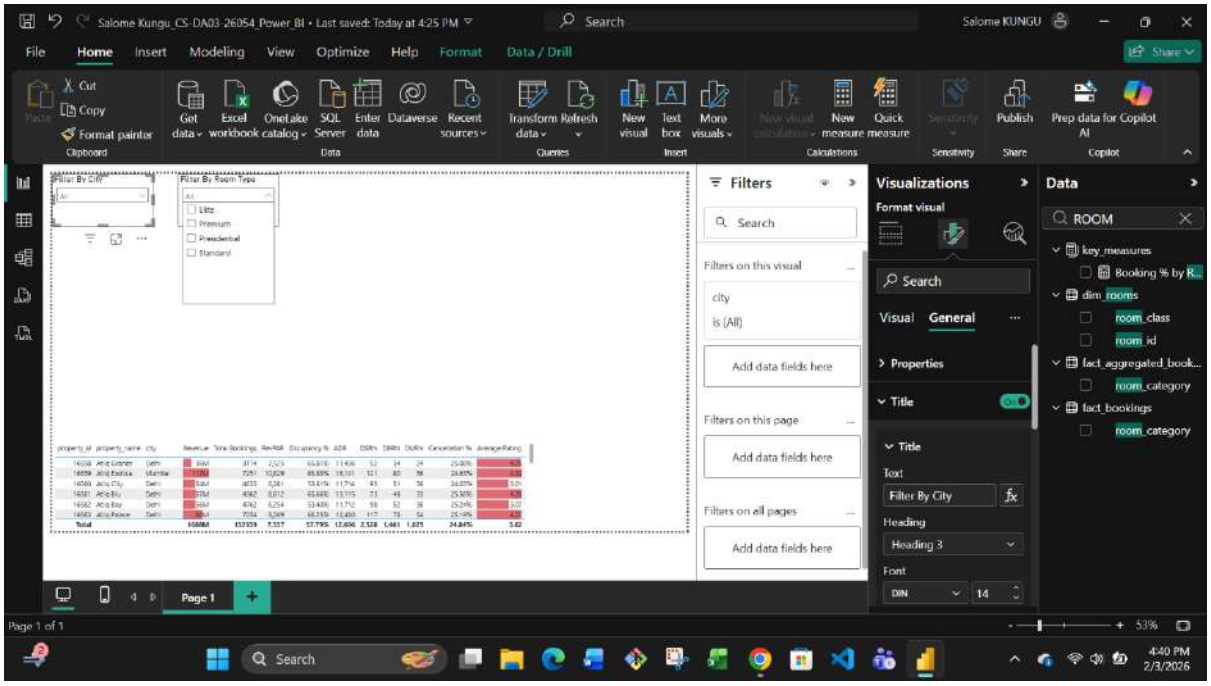
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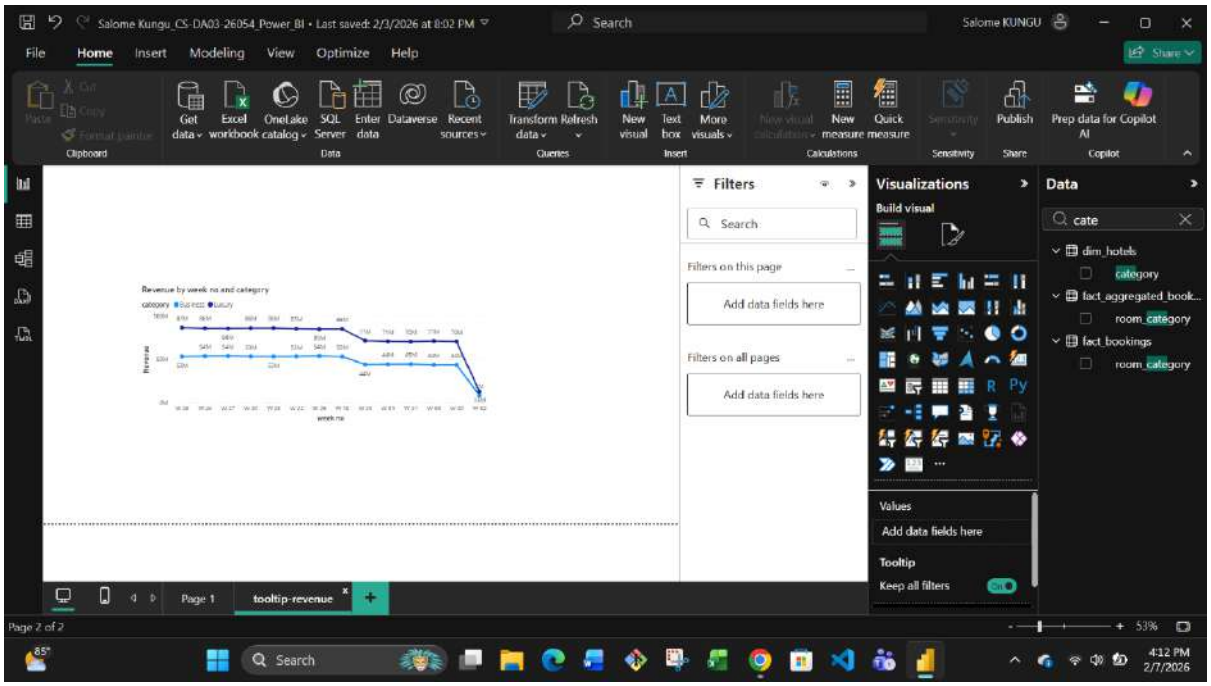
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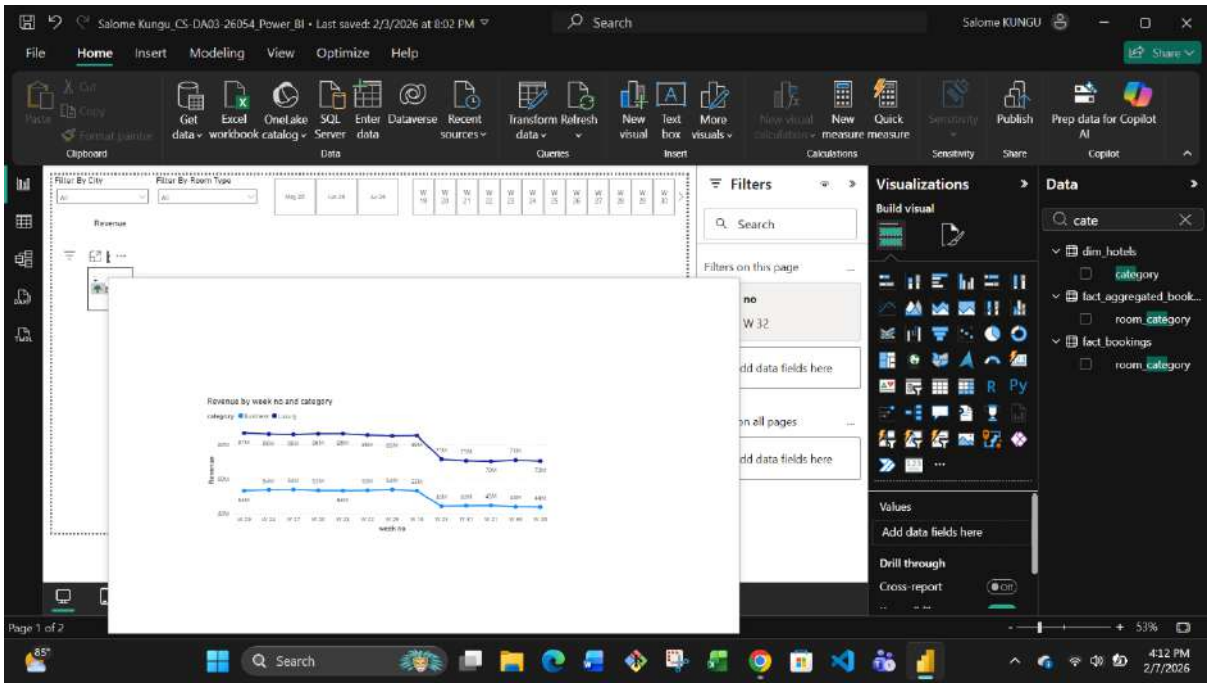
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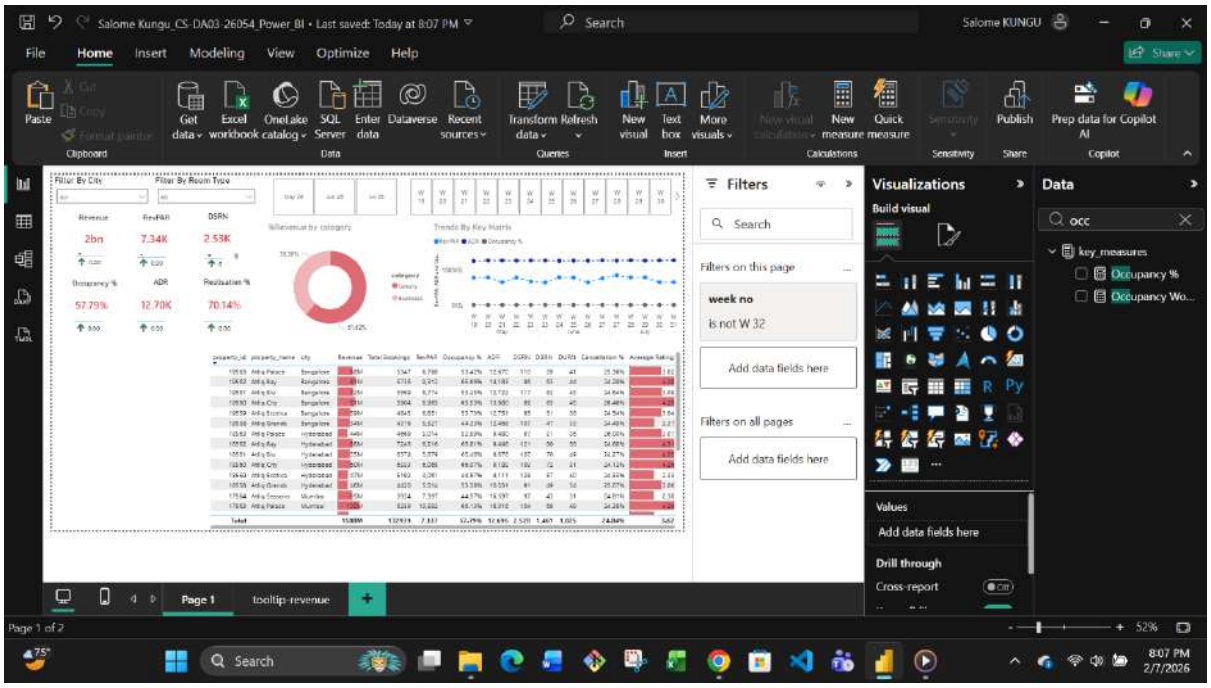
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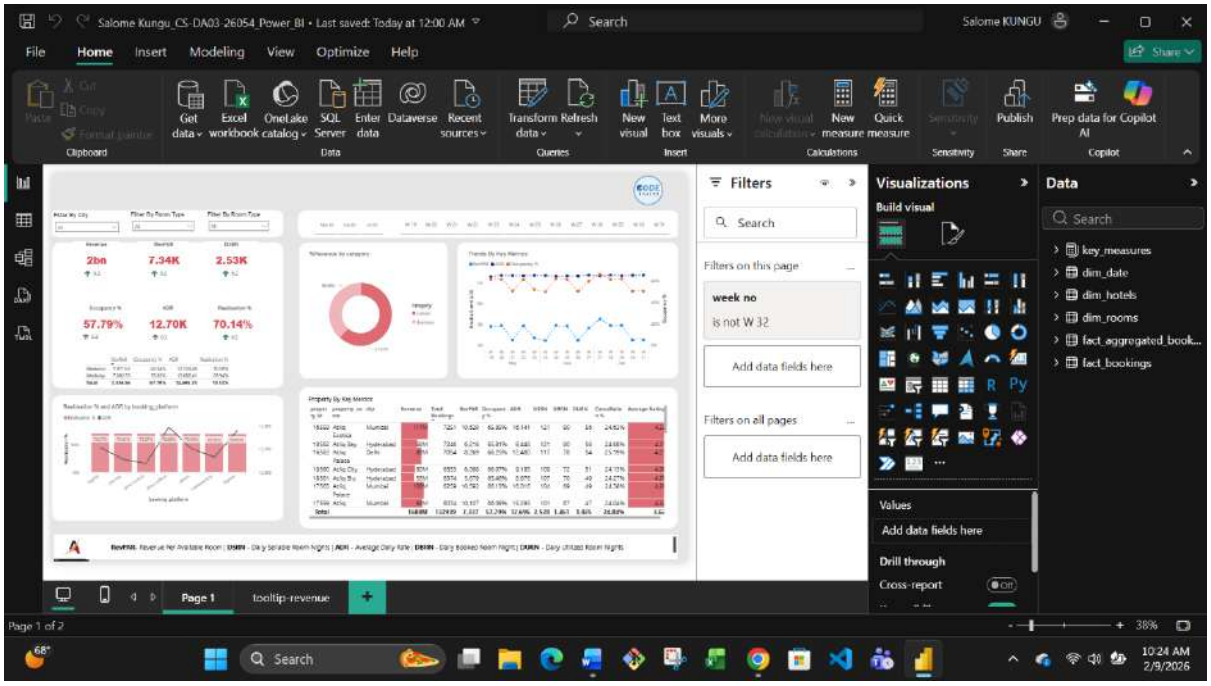
Page 1 of 2



Task 5. Published Report View

Saving and sharing the final Power BI report.
What: The completed Power BI report was published, saved as a PowerPoint file, and both files were uploaded to Google Drive with a shared access link.
Why: This ensured easy access, sharing, and submission of the project for review and portfolio purposes.